A892xx Single-Shunt Sensorless FOC Motor Control Example Code Application Note

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1 Introduction

As increasing numbers of motor control applications place higher demands on energy efficiency (such as automotive electrification, data centers, and industrial automation), motor control technologies are required to achieve an optimal balance among high efficiency, low acoustic noise, low cost, and high reliability. Traditional trapezoidal control can no longer meet these requirements, and Field-Oriented Control (FOC) has become the mainstream solution.

The core principle of FOC is to utilize coordinate transformation to convert the motor's complex three-phase AC currents into independent control variables similar to those of a DC motor, thereby achieving complete decoupling of flux and torque control. Its advantage lies in improving the system's dynamic response, enabling BLDC and PMSM motors to maintain high efficiency across the entire speed range.

Core Advantages of Allegro SoC A892xx Single-Shunt Sensorless FOC Motor Control Solution：

- **High Integration**: The A892xx series integrates an ARM Cortex-M4 microcontroller (MCU), a three-phase motor gate driver (GDU), and a high-precision differential current sensing operational amplifier. This enables efficient implementation of single-shunt sensorless FOC motor control.
- **Low Cost**: By utilizing only one current sensing resistor in combination with the SoC architecture, the solution significantly reduces components and PCB size, thereby lowering overall system cost.
- **Rotor Position Observer**: A sliding mode observer (SMO) algorithm is employed, minimizing dependency on motor parameters and enhancing system robustness.
- **Three-Phase Current Reconstruction Algorithm**: Provides accurate reconstruction of the motor's three-phase currents.
- **High Efficiency**: On one hand, the adoption of FOC motor control technology ensures high operational efficiency. On the other hand, Allegro offers a comprehensive tool chain that streamlines the debugging process, further improving development efficiency.

This Application Note Covers

- Overview of the Allegro A892xx single-shunt sensorless FOC motor control system.

- Principles of key control algorithms, including the Sliding Mode Observer (SMO), SVPWM algorithm, and single-shunt-based 3-phase current reconstruction algorithm.
- Example code, debugging tools, and tuning steps.

Through this document, users will learn how to leverage the Allegro A892xx single-shunt sensorless FOC motor control solution and its associated debugging tools to achieve high-performance motor control while reducing system cost, thereby meeting modern application requirements for energy efficiency, reliability, and system integration. Please note that this document is intended solely to explain the provided example code. The example code is for demonstration and evaluation purposes only and is not intended for use in production or commercial applications.

2 System Architecture Overview

2.1 System Block Diagram

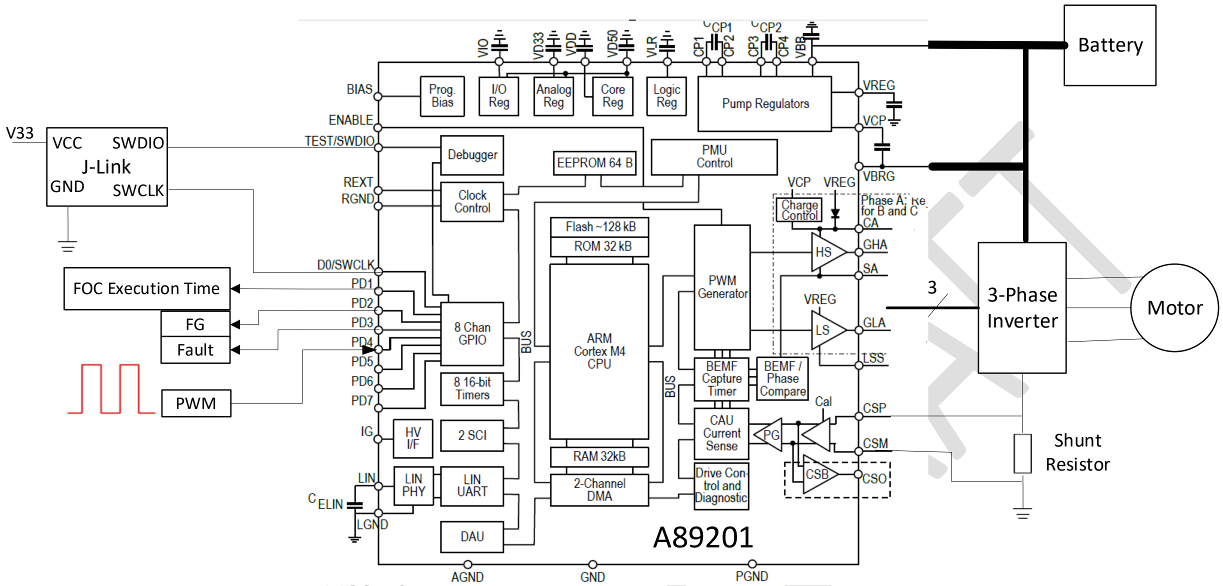


Figure 1 A89201 Single-Shunt Sensorless FOC Motor Control Example System Block Diagram

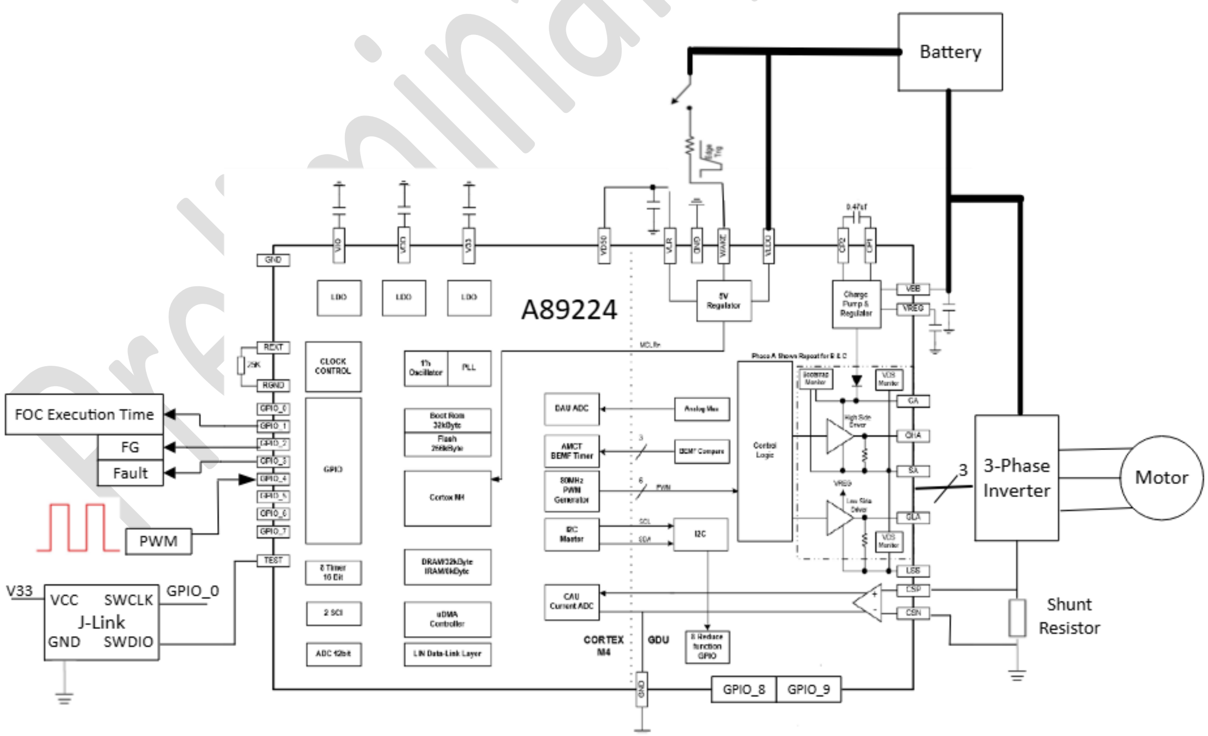


Figure 2 A89224 Single-shunt Sensorless FOC Motor Control Example System Block Diagram

This FOC example code is primarily developed based on the Allegro A89201 and A89224 SoCs, enabling single-shunt sensorless FOC for motor control. The motor startup process can be broadly divided into three phases:

1) DC alignment
2) open-loop forced rotation
3) closed-loop speed control

2.2 Key Features

- **SMO + PLL**: The system estimates the back electromotive force (BEMF) using a sliding mode observer, while motor speed and angle are calculated using a Phase-Locked Loop (PLL) technology.
- **Phase-Shifted Three-Phase Current Reconstruction**: Allegro's phase-shifted current reconstruction algorithm is used to obtain critical current feedback for precise control.
- **Motor Start/Stop and Speed Control via PWM Duty Commands**: Motor operation is controlled through PWM duty cycle instructions. Speed range, acceleration, and deceleration profiles can be configured based on specific application requirements.
- **Comprehensive Fault Protection Mechanisms**: Includes software over current protection, hardware-level VDS over current protection, power-on reset (POR), system fault detection, over temperature protection, and internal LDO undervoltage monitoring.
- **Diagnostic Feedback for System Evaluation**: Provides essential diagnostic data such as FOC main loop execution time, current rotor electrical frequency (via FG signal), and fault status signal.
- **Real-Time Bus Voltage Compensation via DAU Module**: The DAU module's multi-channel analog output (AOUT) enables real-time sampling of VBB voltage, improving control accuracy, efficiency, and dynamic performance.
- **Debugging Tool Chain**: Supports SEGGER J-Scope for real-time waveform analysis and includes a convenient motor control parameter calculation and conversion tool to help users quickly bring up motor operation.

3 Principles of Key Motor Control Algorithms

3.1 Single-Shunt Sensorless FOC Motor Control Process

1. DC Bus Current Measurement
   Measure the bus current I_{Bus}. Apply the phase current reconstruction algorithm to obtain the three-phase currents I_a, I_b, I_c.
2. Clarke Transformation
   Transform the three-phase currents I_a, I_b, I_c into I_α and I_β. In the stationary reference frame, I_α and I_β are orthogonal to each other, representing the time-varying current components.
3. Park Transformation
   Using the electrical angle calculated in the previous iteration, transform I_α and I_β into I_d and I_q。
4. Speed Loop PI Control
   Compare the speed reference ω_{ref} with the estimated speed ω, the error is fed into the speed PI controller. The PI controller output is the q-axis current reference I_{qref}.
5. Current Loop PI Control
   Compare the current references I_{dref} and I_{qref} with the current feedback I_d and I_q, their errors are fed into the current PI controllers. Typically, $I_{dref} = 0$ to maintain constant rotor flux linkage, except during field-weakening where it is set negative.
   - I_{dref} controls flux linkage
   - I_{qref} controls motor torque

   Under steady-state conditions, I_d and I_q approach constant values.
   The PI controllers output voltage commands V_d and V_q, which represent the voltage vector to be applied to the motor stator windings.
6. Angle and Speed Estimation
   V_α, V_β, I_α, and I_β are fed into the SMO with PLL to estimate the new rotor angle and speed. The updated angle determines the position of the voltage vector for the next FOC PWM cycle.
7. Inverse Park Transformation
   Transform V_d and V_q back into V_α and V_β.
8. Inverse Clarke Transformation and SVPWM
   Convert V_α and V_β into three-phase voltages V_a, V_b, and V_c. Compute and update the new PWM duty cycles and determine the next two ADC sampling trigger timing for bus current measurement.

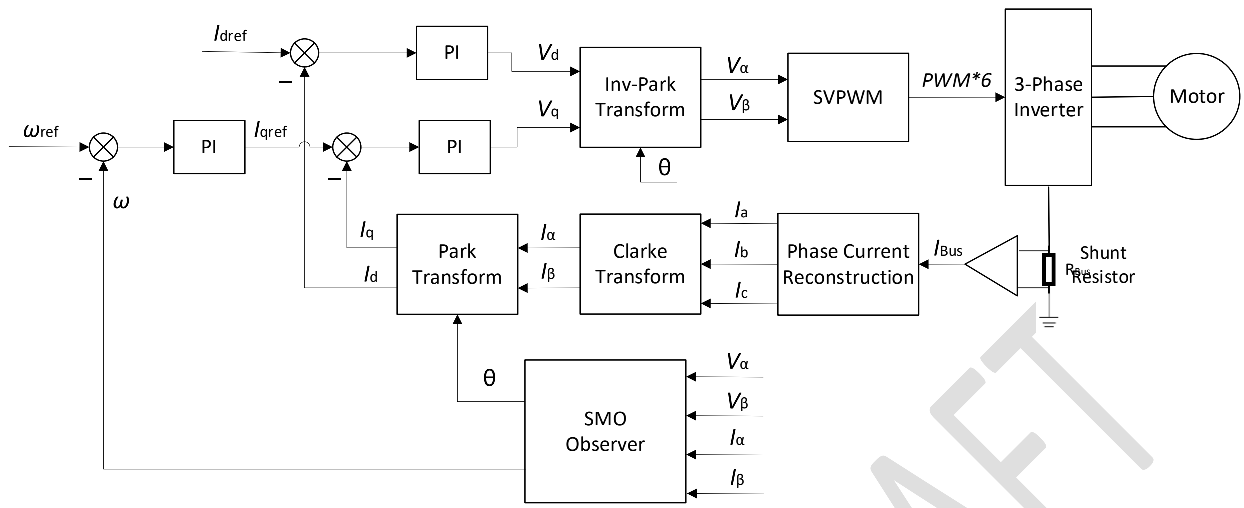


Figure 3 Single-Shunt Sensorless FOC Motor Control Diagram

3.2 Sliding Mode Observer

The mathematical model of a three-phase permanent magnet synchronous motor is expressed as follows:

$$v_s = Ri_s + L\frac{di_s}{dt} + e_s$$

Where:

v_s——stator voltage vector

R——stator winding resistance

i_s——stator current vector

L——stator winding inductance

e_s——back electromotive force (BEMF) vector

The continuous-time current equation derived from the above model is:

$$\frac{di_s}{dt} = -\frac{R}{L}i_s + \frac{1}{L}(v_s - e_s)$$

In digital control systems, the discrete form of the equation is expressed as:

$$i_s(n) = \left(1 - T_s\frac{R}{L}\right)i_s(n-1) + \frac{T_s}{L}\left(v_s(n-1) - e_s(n-1)\right)$$

Where:

T_s——PWM switching period

By applying Clarke transformation, the discrete current equations in the stationary α-β frame are:

$$i_\alpha(n) = \left(1 - T_s\frac{R}{L}\right) i_\alpha(n-1) + \frac{T_s}{L}\left(v_\alpha(n-1) - e_\alpha(n-1)\right)$$

$$i_\beta(n) = \left(1 - T_s\frac{R}{L}\right) i_\beta(n-1) + \frac{T_s}{L}\left(v_\beta(n-1) - e_\beta(n-1)\right)$$

The essence of the SMO is to construct a sliding surface based on the current estimation error and introduce a sign function or saturation function to achieve robust correction.

Since the motor back-EMF contains rotor position information but cannot be directly measured, a correction term $Z(n)$ (implemented as a sign or saturation function) is introduced to replace the back-EMF term $e_s(n)$. This leads to the formulation of a new motor mathematical model, namely the SMO current estimation model:

$$i_\alpha^*(n) = \left(1 - T_s\frac{R}{L}\right) i_\alpha^*(n-1) + \frac{T_s}{L}\left(v_\alpha(n-1) - Z_\alpha(n-1)\right)$$

$$i_\beta^*(n) = \left(1 - T_s\frac{R}{L}\right) i_\beta^*(n-1) + \frac{T_s}{L}\left(v_\beta(n-1) - Z_\beta(n-1)\right)$$

By normalizing the above equations into per-unit (pu) form, the discrete current estimation model becomes:

$$i_{\alpha\_pu}^*(n) = \left(1 - kR_{pu}\right) i_{\alpha\_pu}^*(n-1) + k\left(v_{\alpha\_pu}(n-1) - Z_{\alpha\_pu}(n-1)\right)$$

$$i_{\beta\_pu}^*(n) = \left(1 - kR_{pu}\right) i_{\beta\_pu}^*(n-1) + k\left(v_{\beta\_pu}(n-1) - Z_{\beta\_pu}(n-1)\right)$$

Where:

$$R_{pu} = \frac{R}{Z_{base}}$$

$$L_{pu} = \frac{L}{L_{base}}$$

$$k = \frac{T_s\omega_{base}}{L_{pu}}$$

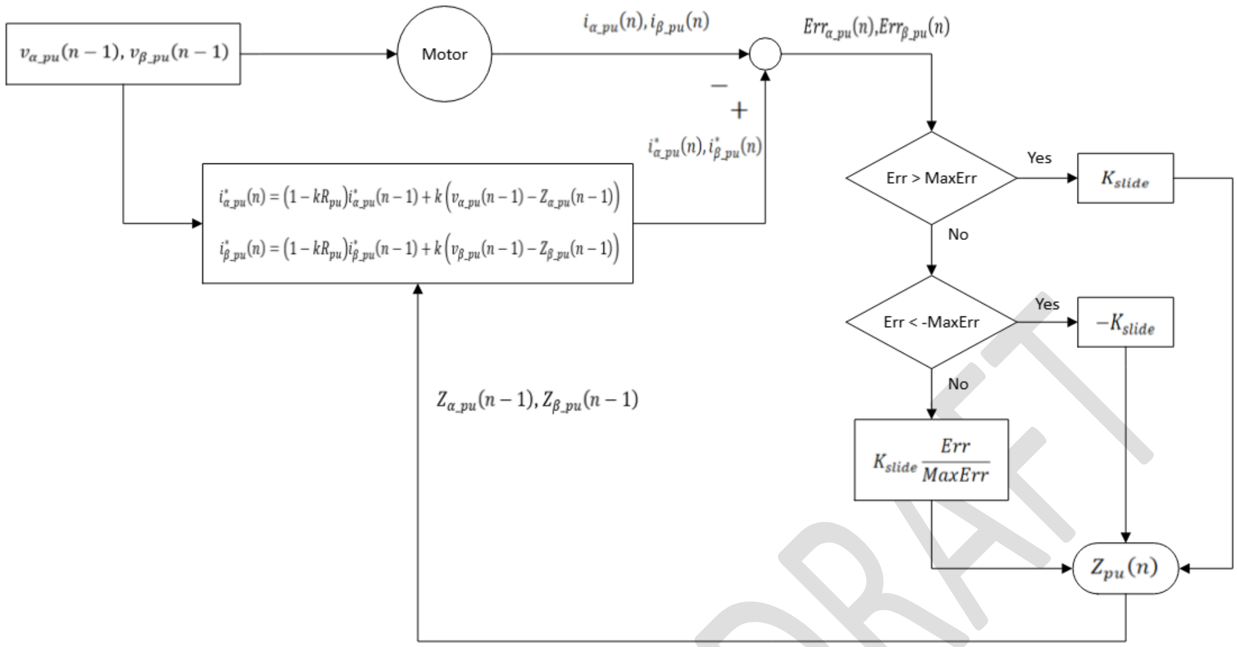


Figure 4 Principles Diagram of the SMO

When the difference between the estimated current and the measured current approaches zero, the constructed motor model can be considered approximately consistent with the actual motor model. By applying a first-order digital low-pass filter, the back-EMF components e_α and e_β can be extracted from the correction term $Z(n)$, which is implemented using a sign or saturation function.

From the e_α and e_β signals, the rotor position and speed information of the motor can be obtained.

3.3 Phase-Locked Loop (PLL) Technology

In steady-state operation, the ideal condition is that the d-axis component of the BEMF is zero ($e_d = 0$). This is because the air-gap flux is mainly produced by the permanent magnets and the d-axis armature reaction flux is zero at this condition.

Based on this principle, a Phase-Locked Loop (PLL) can be used to estimate the motor angular velocity ω. By taking e_d as the error input of the PLL PI controller and driven it to zero in steady state ($e_d \to 0$), consequently, the angular velocity can be obtained.

By integrating the angular velocity ω, the rotor position angle θ can be determined:

$$\theta = \int \omega \, dt$$

Where:

θ——rotor position angle

ω——angular velocity

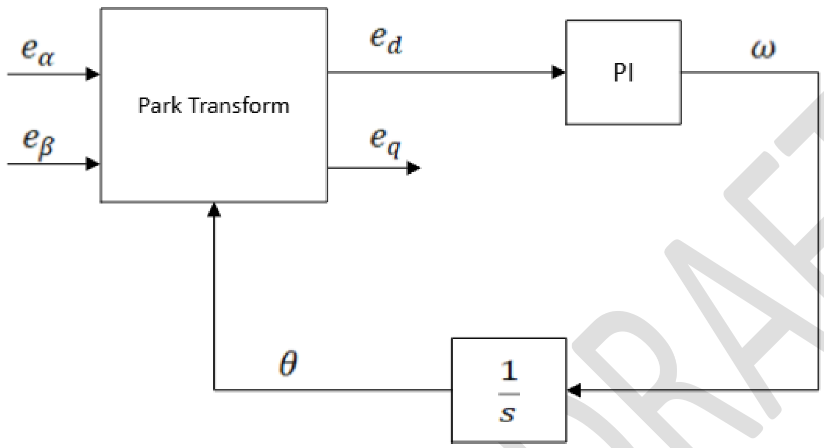
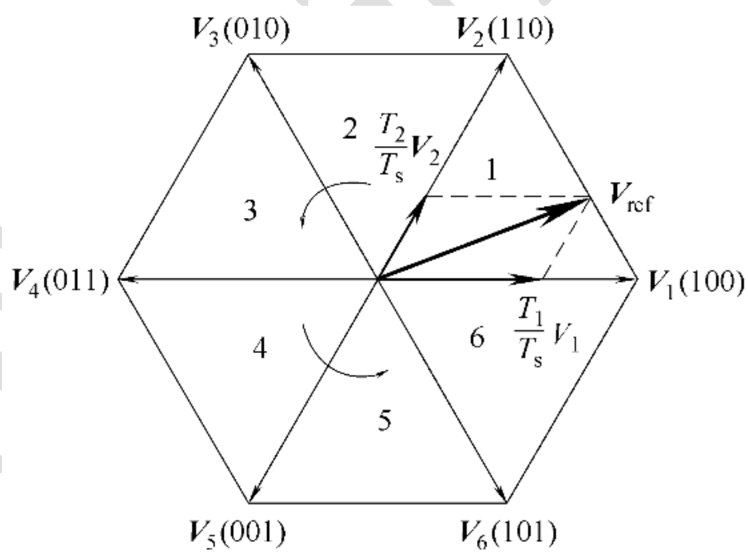


Figure 5 Principle Diagram of the PLL

3.4 Difference Value SVPWM



A three-phase full-bridge inverter circuit consists of six power switching devices, resulting in a total of eight switching states. Among them, 000 and 111 correspond to zero voltage vectors, while the remaining six effective basic voltage vectors are $V_1(100)$, $V_2(110)$, $V_3(010)$, $V_4(011)$, $V_5(001)$, $V_6(101)$. These six vectors evenly divide the 360°

voltage space into six sectors. Any reference vector V_{ref} within this space can be expressed as a linear combination of the six basic voltage vectors and the zero voltage vectors.

$$V_{ref} = \frac{T_k}{T_s} V_k + \frac{T_{k+1}}{T_s} V_{k+1}$$

Where:

V_k, V_{k+1}——two adjacent basic vectors within the same sector

T_k, T_{k+1}——active time of $V_k \ and \ V_{k+1}$

T_s——PWM switching period

In conventional Space Vector PWM (SVPWM), the values of T_k and T_{k+1} can be calculated by the following equations:

$$\begin{cases} T_k = \dfrac{T_s}{V_{dc}} \left|V_{ref}\right| \left(\cos\theta - \dfrac{\sin\theta}{\sqrt{3}}\right) \\ T_{k+1} = \dfrac{2T_s}{\sqrt{3}V_{dc}} \left|V_{ref}\right| \sin\theta \end{cases}$$

Equation 1

Since the reference voltage vector | V_{ref} | is $\frac{3}{2}$ times the phase voltage, the three-phase voltages can be expressed as:

$$\begin{cases} V_a = \dfrac{2}{3} \left|V_{ref}\right| \cos\theta \\ V_b = \dfrac{2}{3} \left|V_{ref}\right| \left(\cos\theta - \dfrac{2}{3}\pi\right) \\ V_c = \dfrac{2}{3} \left|V_{ref}\right| \left(\cos\theta + \dfrac{2}{3}\pi\right) \end{cases}$$

Equation 2

Taking Sector 1 as an example, where $V_a > V_b > V_c$, from the above equations it can be derived that the voltage difference between phase A and phase B is:

$$V_a - V_b = \left|V_{ref}\right| \left(\cos\theta - \frac{\sin\theta}{\sqrt{3}}\right)$$

Equation 3

After rearrangement, we obtain:

$$\left(\cos\theta - \frac{\sin\theta}{\sqrt{3}}\right) = \frac{V_a - V_b}{|V_{ref}|}$$

Equation 4

Similarly, the voltage difference between phase B and phase C can be obtained as:

$$V_b - V_c = \frac{2}{\sqrt{3}}|V_{ref}|\sin\theta$$

Equation 5

After rearrangement, we obtain:

$$\sin\theta = \frac{\sqrt{3}}{2|V_{ref}|}(V_b - V_c)$$

Equation 6

By substituting Equation 4 and Equation 6 into Equation 1, we obtain:

$$\begin{cases} T_1 = \dfrac{T_s}{V_{dc}}(V_a - V_b) \\ T_2 = \dfrac{T_s}{V_{dc}}(V_b - V_c) \end{cases}$$

Where：

T_1， T_2——the active time of the two adjacent basic voltage vectors V_1 and V_2 in Sector 1.

From the above equations, it can be seen that the active time of adjacent vectors is proportional to the voltage difference between the corresponding phases. By analogy, the same principle applies to other sectors, and a general expression can be derived. Assuming the three phase voltages are arranged in descending order as $V_{max} > V_{mid} > V_{min}$, the active time of the adjacent voltage vectors are given by:

$$\begin{cases} T_k = \dfrac{T_s}{V_{dc}}(V_{max} - V_{mid}) \\ T_{k+1} = \dfrac{T_s}{V_{dc}}(V_{mid} - V_{min}) \end{cases}$$

Based on the magnitude relationship of the three-phase voltages, the active time of the two adjacent basic voltage vectors within the sector can be calculated.

Table 1 Sector Determination Method

Sector	**Angle θ**	***Magnitude Relationship of V_a, V_b, V_c***
1	$0 \leq \theta < \frac{\pi}{3}$	$V_a > V_b > V_c$
2	$\frac{\pi}{3} \leq \theta < \frac{2\pi}{3}$	$V_b > V_a > V_c$
3	$\frac{2\pi}{3} \leq \theta < \pi$	$V_b > V_c > V_a$
4	$\pi \leq \theta < \frac{4\pi}{3}$	$V_c > V_b > V_a$
5	$\frac{4\pi}{3} \leq \theta < \frac{5\pi}{3}$	$V_c > V_a > V_b$
6	$\frac{5\pi}{3} \leq \theta < 2\pi$	$V_a > V_c > V_b$

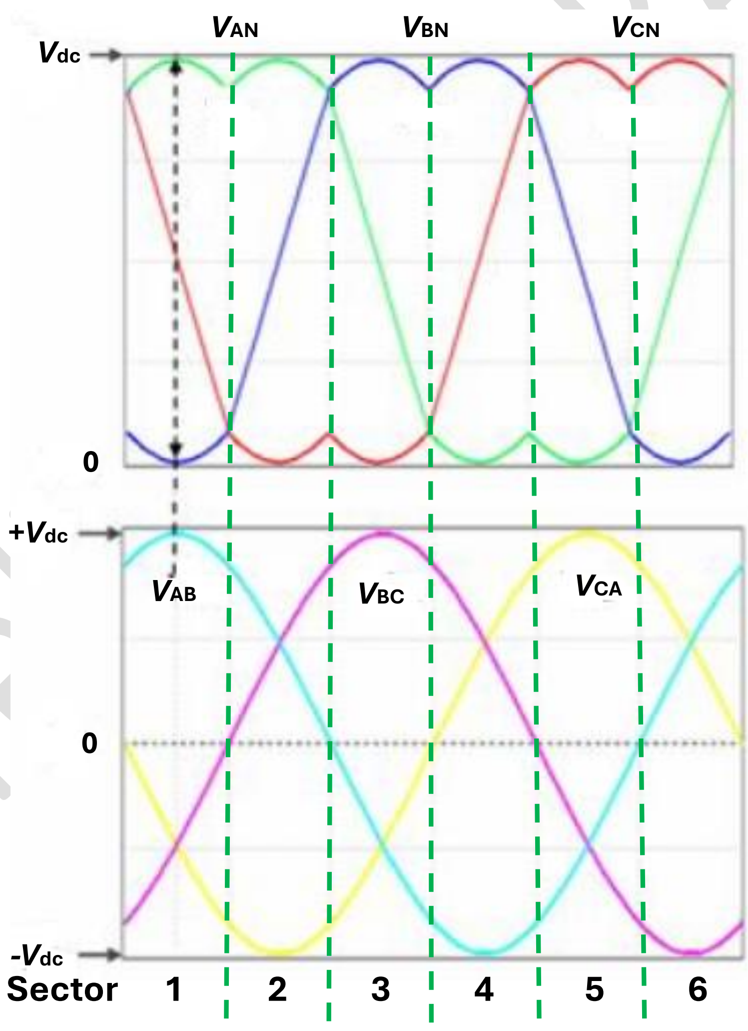


Figure 6 SVPWM Modulation: Phase Voltage, Line Voltage, and Sector

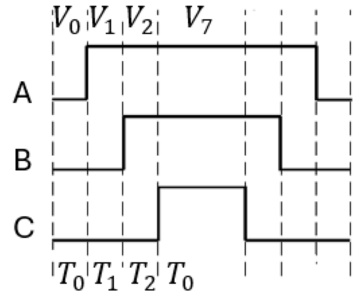


Figure 7 Illustration of the seven-segment pulse sequence in SVPWM

Taking Sector 1 as an example again, the figure above illustrates the seven-segment pulse sequence of SVPWM. Considering that the FOC example code adopts a center-symmetric PWM mode, only half of the PWM cycle modulation process is analyzed.

The active time of V_0, V_1 and V_2 correspond to T_0, T_1 and T_2, respectively. From the above equations, it can be seen that:

- The voltage difference between phase A and phase B determines T_1.
- The voltage difference between phase B and phase C determines T_2.

The switching time of the upper switches in the three-phase bridge arms, $T_a, T_b \ and \ T_c$, are given by:

$$\begin{cases} T_a = T_0 \\ T_b = T_0 + T_1 \\ T_c = T_0 + T_1 + T_2 \end{cases}$$

The derivations for the remaining sectors are similar and will not be repeated here. Based on the calculated values of $T_a, T_b \ and \ T_c$, the duty cycles of the PWM signals can be determined.

3.5 Phase-Shifted Three-Phase Current Reconstruction

Single-shunt sampling uses only one shunt resistor and a single differential operational amplifier circuit, which significantly reduces system cost. However, it should be noted that in order to obtain valid current information, the SVPWM modulation scheme needs to be modified under certain operating conditions. Such modifications may lead to current distortion.

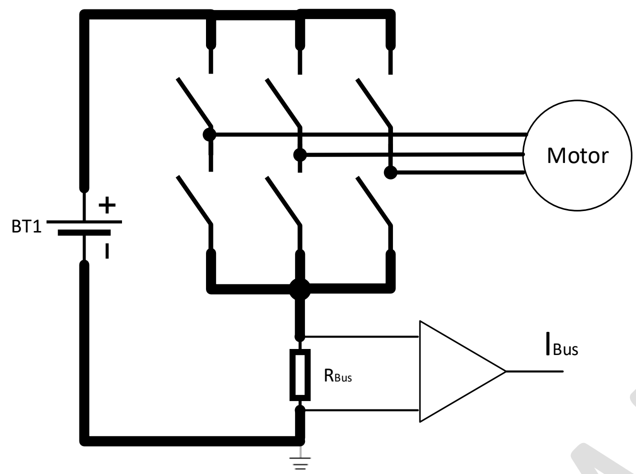


Figure 8 Bus Current Sampling Circuit

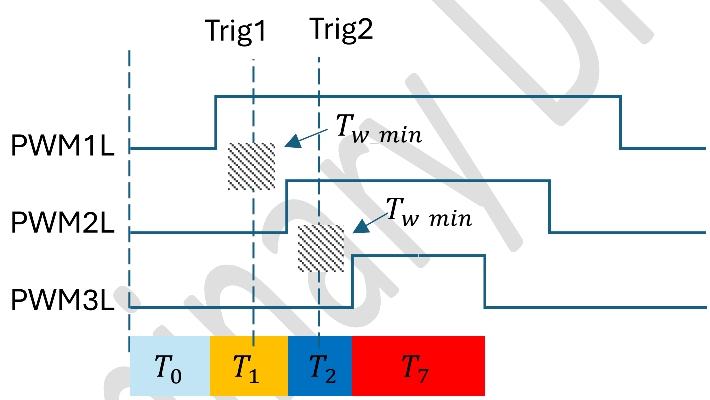


Figure 9 Ideal Sampling Window

For the single-shunt current sampling algorithm, it is essential to ensure that sampling is performed when the lower bridge arm is conducting, so that the bus current flowing through the shunt resistor can be measured. Therefore, the design of the sampling trigger timing and sampling window is particularly critical. Ideally, the active time of T_1 and T_2 should both be greater than the minimum sampling window time T_{w_min}. The value of T_{w_min} depends on:

- PWM frequency. The higher the frequency, the smaller T_{w_min}.
- Dead time.
- Slew rate of the differential amplifier and delay of the RC filter circuit.
- Switching noise at the instant of power device commutation.

It should also be noted that the two ADC sampling trigger instants, Trig1 and Trig2, must avoid the switching noise and current spikes of the power devices, in order to prevent measurement distortion.

The following two situations may cause the effective sampling window to become too narrow within certain PWM cycles, making it impossible to accurately and reliably measure the bus current.

Case 1: When the motor is running under no load or with a very light load, due to the complementary center-aligned PWM mode, the duty cycle of the PWM may approach 50%. In this case, both T_1 and T_2 are smaller than T_{w_min}.

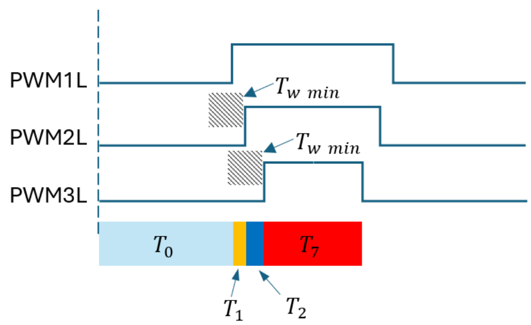


Figure 10 $T_1 < T_{w\_min}$ *and* $T_2 < T_{w\_min}$

Case 2: When the motor is running under heavy load or full load, some PWM duty cycles become similar or equal. In this case, either T_1 or T_2 is smaller than T_{w_min}. The figure below illustrates the case where $T_2 < T_{w\_min}$.

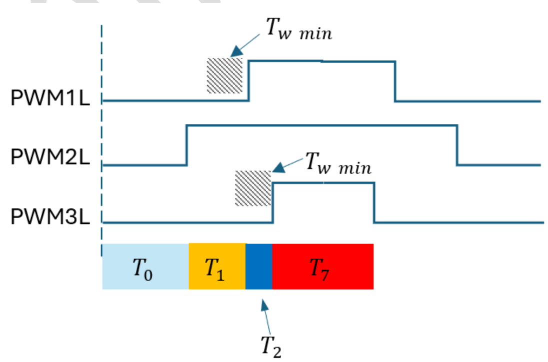


Figure 11 $T_2 < T_{w\_min}$

To address the issue of an excessively narrow sampling window, this FOC example code adopts a phase-shifted PWM duty cycle adjustment strategy to ensure the minimum sampling window time. The specific approach is outlined as follows.

Duty cycle before modification:

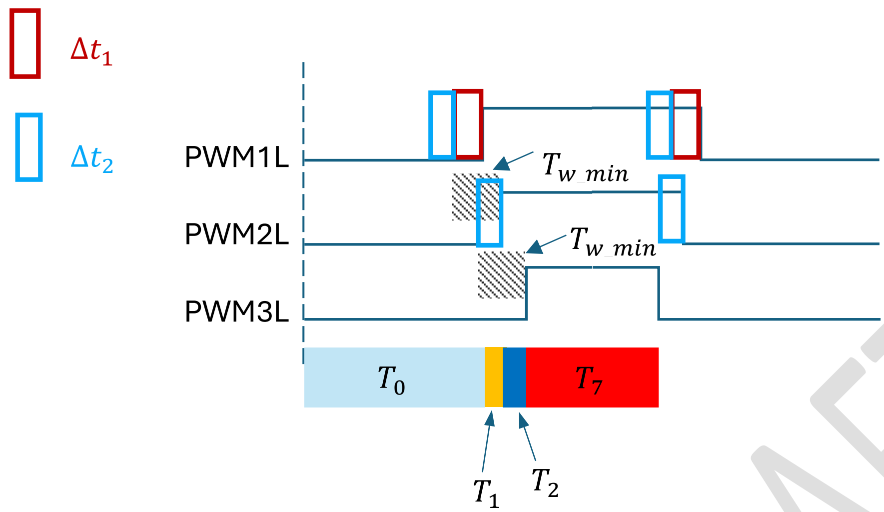


Figure 12 Duty Cycle before Modification

$$\Delta t_1 = T_{w\_min} - T_1$$

$$\Delta t_2 = T_{w\_min} - T_2$$

Where:

$\Delta t_1$and Δt_2 indicates the direction and value of duty cycle adjustment.

Duty cycle after modification：

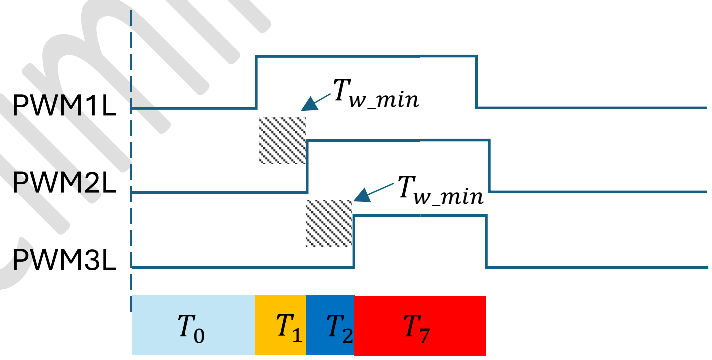


Figure 13 Duty Cycle after Modification

4 Example Code Introduction

4.1 Peripherals Configuration

1. GPIO Configuration

PIN	Mode	Function Description
GPIO1	Push-pull output	Evaluate the FOC main control code execution time
GPIO2	Push-pull output	FG signal output (rotor electrical frequency)
GPIO3	Push-pull output	Fault status signal output
GPIO4	Digital input	PWM generator signal input

2. Timer Configuration

Timer	Mode	Function Description
Timer1	Timing	Periodic 1 ms. Periodically triggers DAU sampling.
Timer2	Timing	Periodic 250 μs. Periodically executes the speed control loop
Timer5	High-time measurement	Measures the duration of the high-level state of the PWM input signal
Timer6	Period measurement	Measures the entire period of the PWM input signal
Timer7	Timing	Periodic 5 ms or 200 ms. Periodically executes PWM input signal parsing, converting duty cycle into speed reference command.

3. Interrupt Priority Configuration

Interrupt	Priority	Function Description
CAU	0	Executes FOC main control code
Timer2	1	Executes speed control loop
Timer7	2	Converts PWM input signal into speed reference command
DAU	3	Performs bus voltage sampling

4.2 Flowchart

1. **Main.c Flowchart**

Main.c overall structure

1) Initialization Phase

During the initialization phase of the main.c, the following tasks must be completed:

- **Interrupt priority configuration** to ensure real-time execution of critical control tasks.
- **FOC control parameter initialization**, including current loop, speed loop, motor control state machine, and coordinate transformation.
- **User parameter loading**, covering normalization values, limiting strategies, and startup configuration.
- **PWM input signal parsing function initialization**, used to obtain speed reference command.
- **Peripheral initialization**, including CAU, DAU, GDU, GPIO, PGU, and GTU.

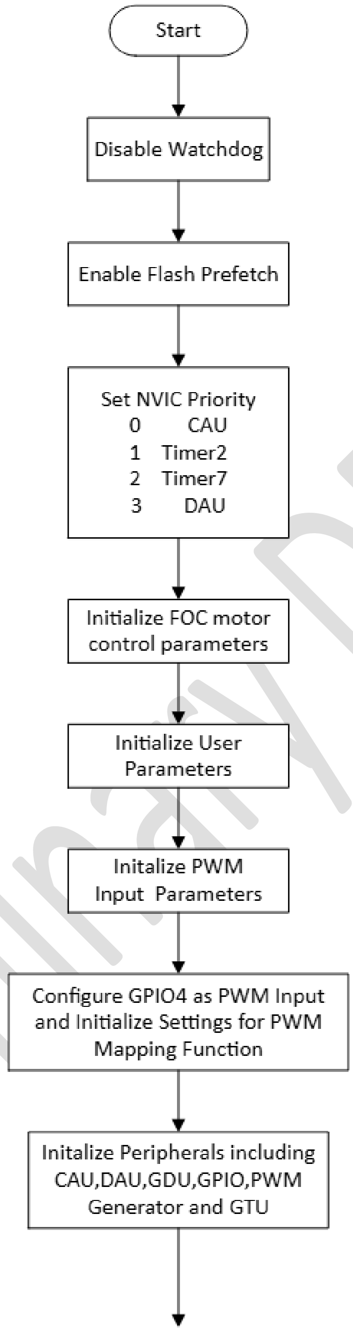
2) Main loop (while) execution phase

Once entering the main loop, the following tasks are performed:

- **Fault detection and handling**: continuously monitor system status. In case of a fault, immediately stop motor operation and output fault status signals to ensure system safety.
- **Normal operation logic**: when a non-zero speed command is received, start the motor; otherwise, stop the motor.
- **FG signal generation**: outputs FG signal based on rotor angle information, used for real-time speed feedback.

3) Interrupt service routine execution phase

- **CAU ISR**: executes the core FOC control program, triggered once per PWM period. PWM period can be set by the user.
- **Timer2 ISR**: executes the speed control loop task, execution frequency 4 kHz.
- **Timer7 ISR**: parses PWM input duty cycle and converts it into a speed command, execution frequency selectable as 5 Hz or 200 Hz.
- **DAU ISR**: uses the internal multi-channel analog AOUT module to sample bus voltage, applied for FOC bus voltage compensation.


Start
Disable Watchdog
Enable Flash Prefetch
Set NVIC Priority
0 CAU
1 Timer2
2 Timer7
3 DAU
Initialize FOC motor control parameters
Initialize User Parameters
Initalize PWM Input Parameters
Configure GPIO4 as PWM Input and Initialize Settings for PWM Mapping Function
Initalize Peripherals including CAU,DAU,GDU,GPIO,PWM Generator and GTU

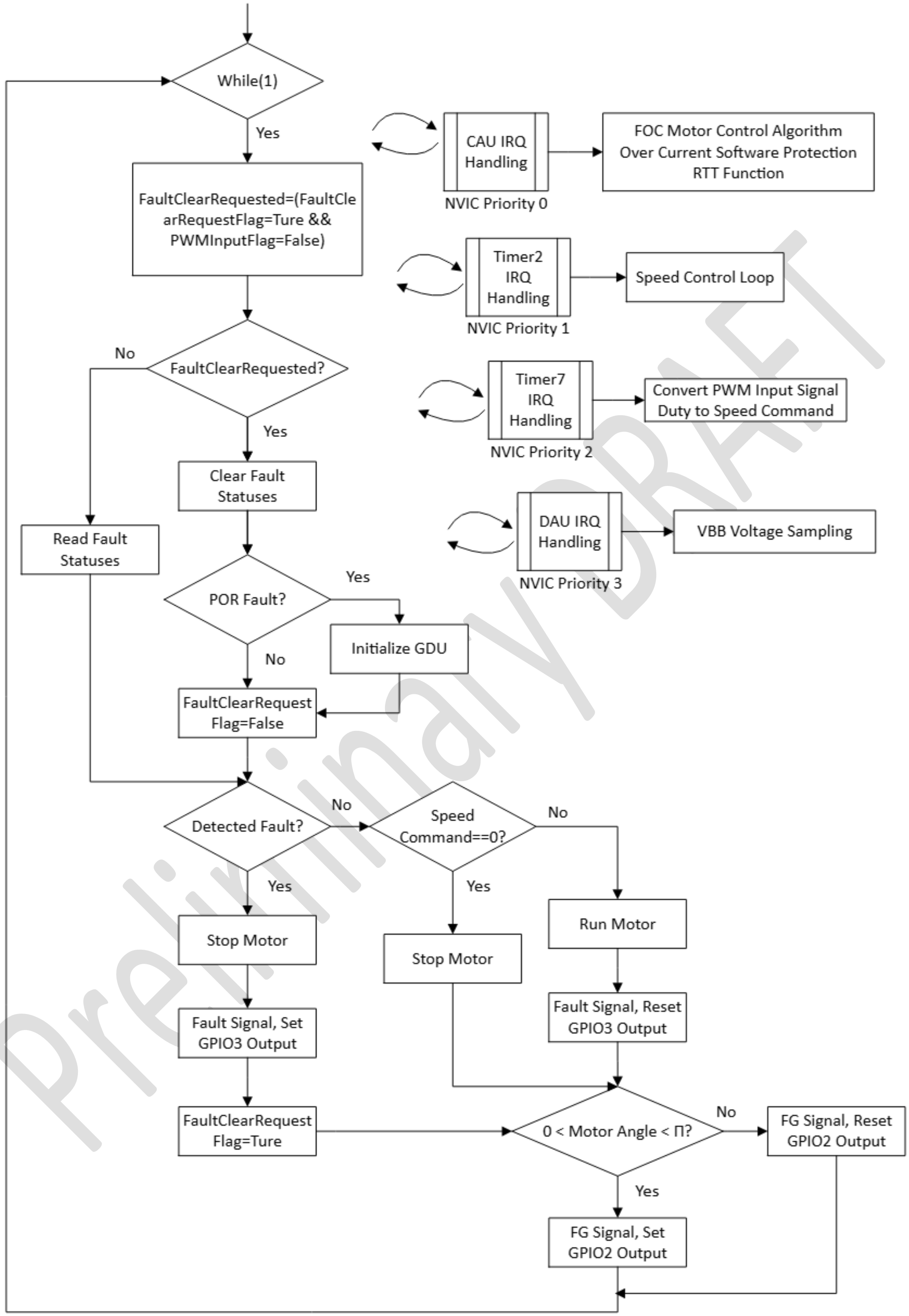


Figure 14 Main.c Flowchart

2. CAU ISR Flowchart

Function: Executed once per PWM period to perform the core FOC control tasks and evaluate execution time.

Execution Steps:

1) **Set GPIO1 High**
   Used to measure the execution time of the FOC core algorithm.

2) **Execute Motor Control State Machine**
   The state machine includes the following stages:

   a. **Idle state:** Motor does not get started, system remains in safe and idle state. CAU samples differential amplifier output offsets and applies digital filtering.

   b. **DC alignment:** Apply directional current to align rotor position with reference.

   c. **Open-loop forced acceleration:** Accelerate the motor to the closed-loop switching speed using fixed frequency.

   d. **Open-loop to closed-loop transition:** Smoothly switch to closed-loop control.

   e. **Closed-loop speed control:** Achieve high-precision speed regulation based on the FOC algorithm.

3) **Sample Bus Current and Reconstruct Three-Phase Currents**
   Use phase-shifted reconstruction algorithm to obtain three-phase current information.

4) **Software Overcurrent Protection**
   Detect in real time according to user-configured protection thresholds and trigger protective actions to ensure system safety.

5) **Coordinate Transformation and Decoupling Control**
   Perform Clarke and Park transformations to decouple d-q axis current components, enabling independent control of motor torque and excitation.

6) **Update ADC Sampling Trigger Timing**
   Ensure current flowing through the single shunt resistor is sampled when the lower bridge MOSFET is conducting.

7) **Update PWM Duty Cycle**
   Compute and update PWM duty cycles to drive the motor.

8) **Optional RTT Function**
   User can enable/disable RTT functionality for debugging and data logging.

9) **Clear Interrupt Flag**
   Prepare for the next interrupt request.

10) **Set GPIO1 Low**
   Marks the end of the FOC core task and completes execution time measurement.

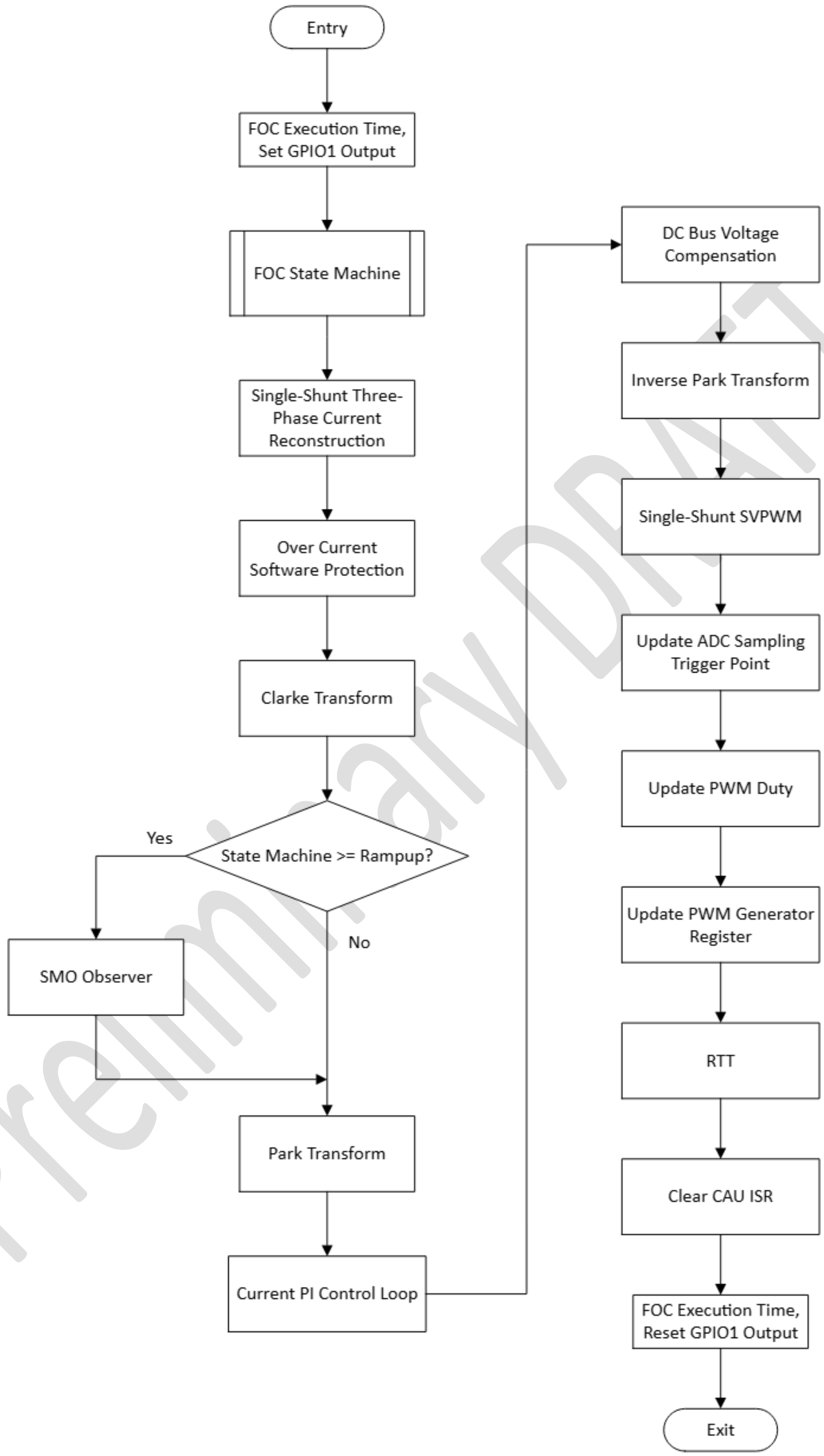


Figure 15 CAU ISR Flowchart

3. Motor Control State Machine Flowchart

1) Idle State

- **Trigger condition:** Stop command received or over current fault detected.
- **Actions:**
  - Force transition to idle state, motor stops.
  - Reset FOC-related control parameters.
  - CAU continuously samples differential amplifier offsets and applies digital filtering to ensure accuracy of current measurements.
- **Waiting:** Start command.

2) Align State

- **Trigger condition:** Start command received.
- **Actions:**
  - Set I_{dref} and I_{qref} reference current values and alignment angle.
  - Maintain alignment current until alignment time expires, ensuring rotor position is aligned with reference.

3) Ramp-up State

- **Actions:**
  - According to the configured ramp-up time, accelerate motor to the closed-loop switching speed threshold.
  - Maintain open-loop current control mode to ensure smooth acceleration.

4) Switching State

- **Actions:** Smooth transition completed in two steps:

  (1) **Drop I_{qref} reference value**
  Purpose: Minimize error between open-loop forced angle and actual rotor position angle, avoiding current surge or transition failure when switching to the closed-loop control.

  (2) **Maintain torque closed-loop control mode**
  According to the configured speed loop waiting time, keep the system in

torque closed-loop mode to ensure smooth transition into speed closed-loop control mode.

5) **Closed-Loop Speed State**

- **Actions:**
    - Switch to speed closed-loop control mode. Achieve high-precision speed regulation based on the FOC algorithm.

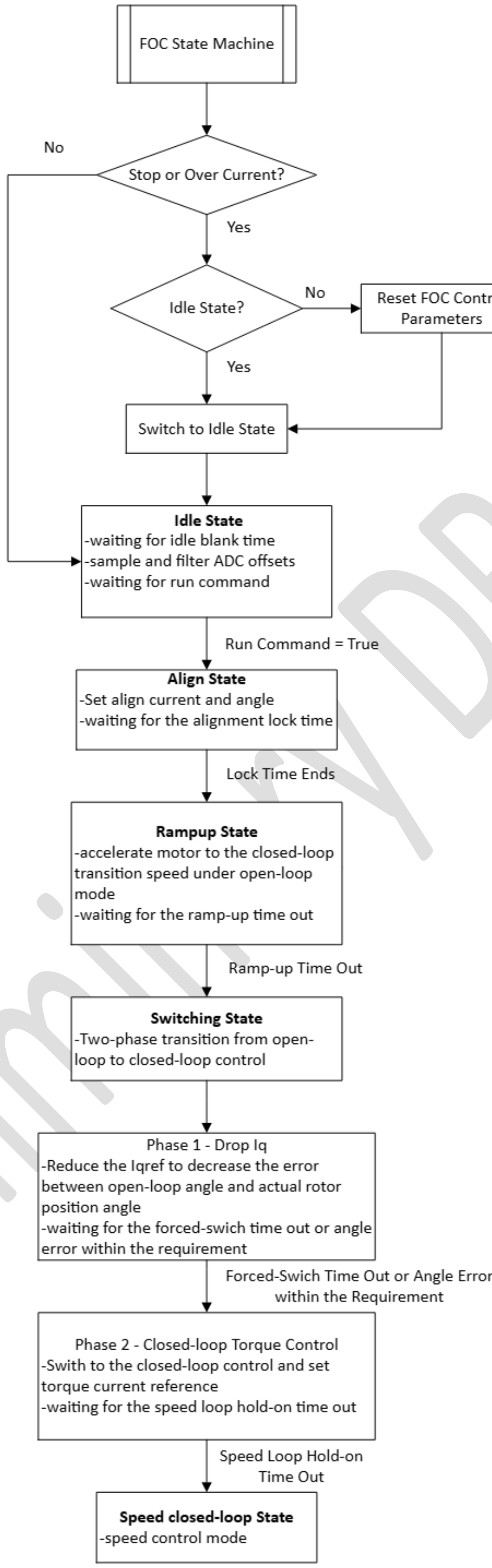


Figure 16 Motor Control State Machine Flowchart

4.3 Speed Variation Curve

To help users better understand this FOC example code, the figure illustrates the motor speed variation throughout the entire startup process.

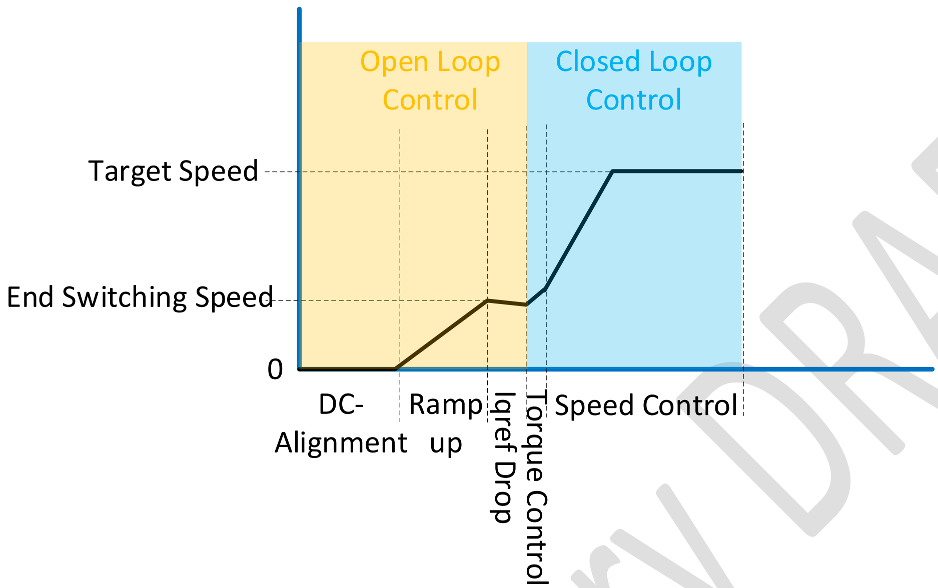


Figure 17 Speed Variation Curve

5 Hardware Introduction

5.1 Allegro A89211/212/224 General Motor Control Demo Board

Figure 18 shows the Allegro A89211/212/224 General Motor Control Demo Board. The schematic, layout, and bill of materials (BOM) for this board are provided as separate documents, which can be downloaded from the Allegro Customer Portal: https://registration.allegromicro.com/login

Figure 18 Allegro A89211/212/224 General Motor Control Demo Board

Table 2 A89211/212/224 General Motor Control Demo Board Specification

Demo Board	**Supply Voltage（V）Min**	**Supply Voltage（V）Max**	**Phase Current（A）Max**
A89211	7	48	30
A89212	7	72	30
A89224 (Automotive qualified)	7	72	30

5.2 Allegro A89211/212/224 General Motor Control Demo Board Wiring Guide

Please connect the DC power supply, PWM signal generator, motor, and ARM debugger correctly as shown in the figure below.

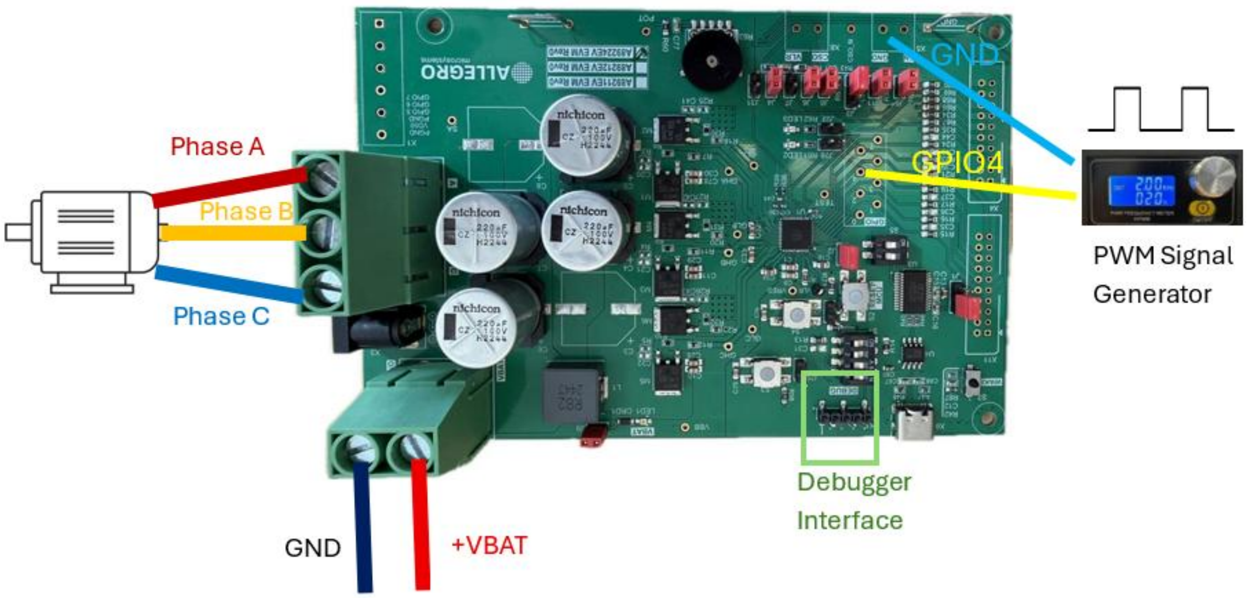


Figure 19 Allegro A89211/212/224 General Motor Control Demo Board Wiring Guide

5.3 Allegro A89201 General Motor Control Demo Board

Figure 20 shows the Allegro A89201 General Motor Control Demo Board. The schematic, layout, and bill of materials (BOM) for this board are provided as separate documents, which can be downloaded from the Allegro Customer Portal: https://registration.allegromicro.com/login

Figure 20 Allegro A89201 General Motor Control Demo Board

Table 3 A89201 General Motor Control Demo Board Specification

Demo Board	Supply Voltage (V) Min	Supply Voltage (V) Max	Phase Current (A) Max
A89201	7	40	40

5.4 Allegro A89201 General Motor Control Demo Board Wiring Guide

Please connect the DC power supply, PWM signal generator, motor, and ARM debugger correctly as shown in the figure below.

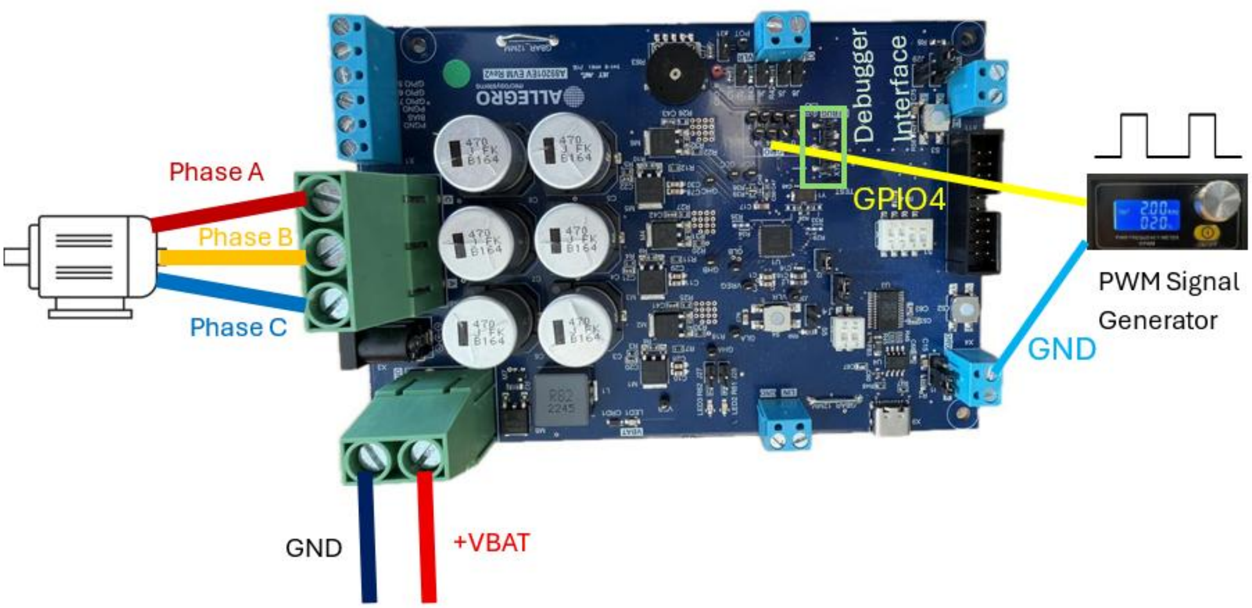


Figure 21 Allegro A89201 General Motor Control Demo Board Wiring Guide

6 Debugging Tools

6.1 Control Parameter Normalization Description

This FOC example code uses fixed-point Q15 format for computation, and all motor control parameters are processed with corresponding per-unitization. For detailed data conversion derivation and definitions, please refer to **UserParameters_Guideline.txt**. The document can be accessed in the following two ways:

1. Load the FOC example code in the Keil embedded development tool. Locate the document at the position shown in the figure below.

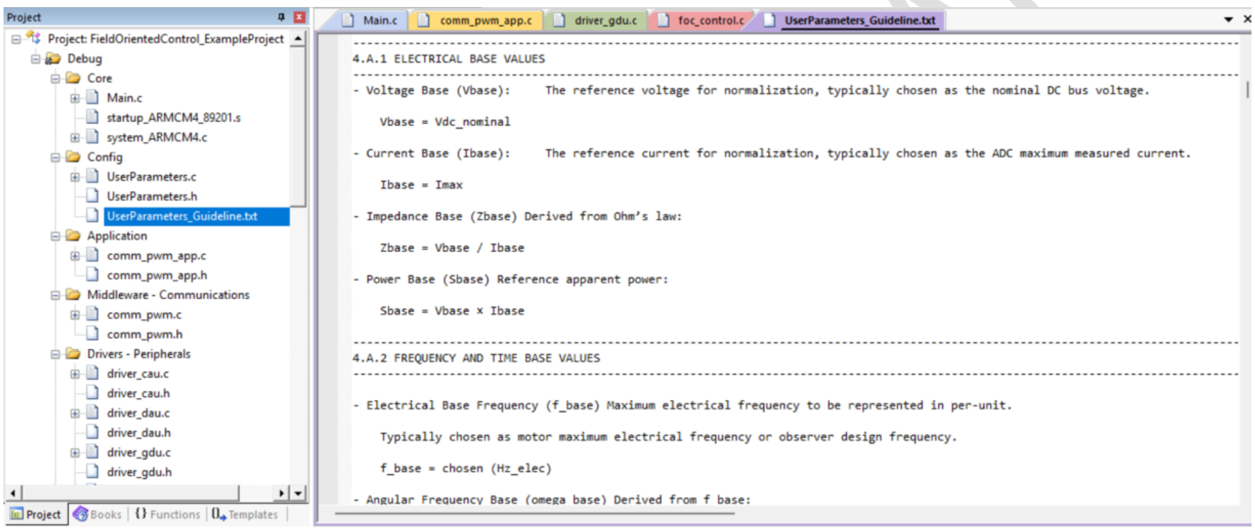


Figure 22 UserParameters_Guideline.txt in the Code

2. Locate the document through the folder path: A892xx_FOC_Example_Code\A892xx_C\A89224\Application\FOC\Config

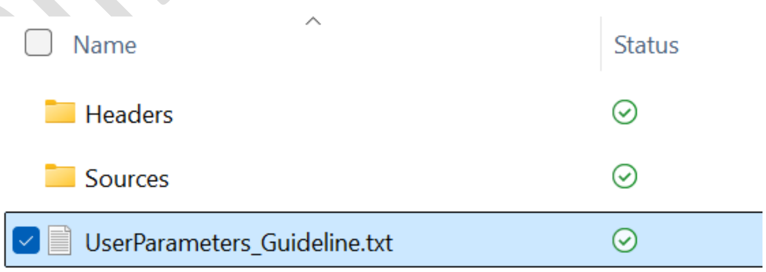


Figure 23 UserParameters_Guideline.txt in the Folder

6.2 Parameter Calculation and Conversion Tool

1. **Introduction**

To facilitate user debugging, Allegro provides a parameter calculation and conversion tool: **A892xx_FOC_Example_Code_Parameters_Calculator.exe**. By simply typing the motor control parameters, users can quickly obtain the per-unit parameters suitable for this FOC example code, eliminating the need for tedious manual calculations.

When launching **A892xx_FOC_Example_Code_Parameters_Calculator.exe**, the interface is displayed as shown below.

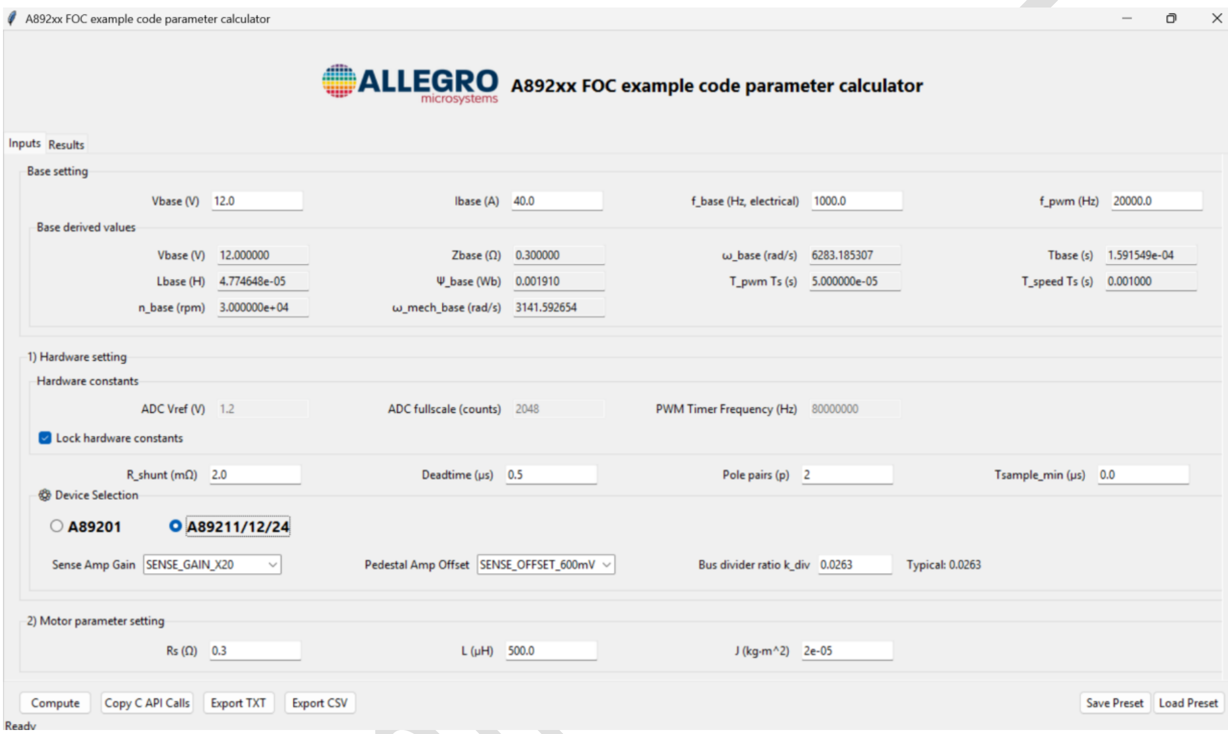
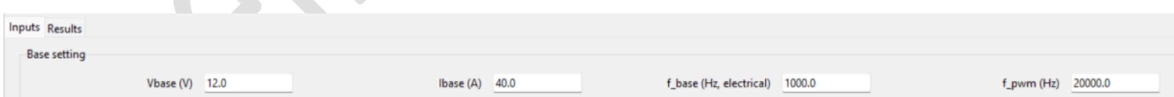


Figure 24 A892xx_FOC_Example_Code_Parameters_Calculator User Interface

2. User Guide

1) Base setting



V_{base}: Maximum system voltage

I_{base}: Maximum system current

f_{base}: Maximum electrical frequency of the motor

f_{PWM}: PWM switching frequency

2) Base derived values

The following base values are automatically calculated from the user-entered parameters and do not require modification.

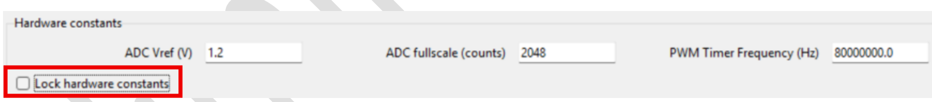
3) Hardware Setting

This section is divided into two parts:

- Part 1: Hardware constants
- Part 2: Device Selection

Part 1: Hardware constants

ADC Vref (ADC reference voltage), **ADC fullscale (counts)** (ADC resolution), and **PWM Timer Frequency** (PWM generator clock frequency) usually do not require modification. If modification is needed, please uncheck **Lock hardware constants**, as shown in the figure below.



Users need to fill in the following four parameters according to the actual application:

R_shunt: Bus current sensing resistor value, unit: mΩ.

- If using the Allegro A89201 General Motor Control Demo Board, set R_shunt = 2.0 mΩ.
- If using the A89211/212/224 General Motor Control Demo Board, set R_shunt = 1.0 mΩ.

Deadtime: Dead time, unit: µs.

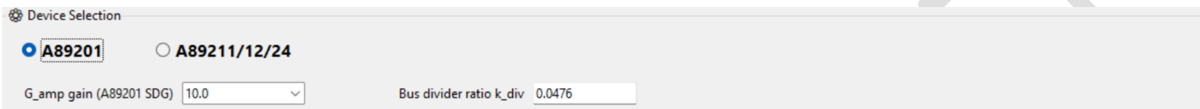
Pole pairs: Number of pole pairs.

Tsample_min： Minimum sampling window time, unit: μs, used for the single-shunt FOC algorithm.

Part 2: Device Selection

Due to differences in the CAU module between the A89201 and the A89211/12/24 series, users need to select the corresponding chip model according to the actual application.

A89201

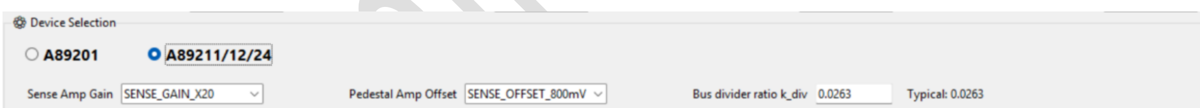


G_amp gain： Differential operational amplifier gain.

Bus divider ratio k_div： Conversion ratio of the bus voltage sampling value. This parameter usually does not require modification. If there is a significant deviation between the sampled value and the actual value, this ratio can be fine-tuned for correction.

Since the effective detection range of the CAU module in the A89201 is −1.2 V to +1.2 V, no additional offset setting is required for the differential operational amplifier output.

A89211/12/24

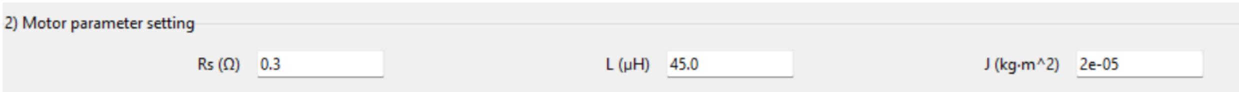


Sense AMP Gain： Differential operational amplifier gain.

Pedestal AMP Offset： Differential operational amplifier output offset value. Since the effective detection range of the CAU module in A89211/12/24 is 0.3 V to 1.2 V, the offset value is generally set to 800 mV.

Bus divider ratio k_div： Conversion ratio of the bus voltage sampling value. This parameter usually does not require modification. If there is a significant deviation between the sampled value and the actual value, this ratio can be fine-tuned for correction.

4) Motor parameter setting



Rs： Motor phase resistance, unit: Ω.

L： Motor phase inductance, unit: μH.

J： Moment of inertia coefficient, unit: kg·m^2.

5) Control processing setting

This section focuses on parameter settings for the four key stages of motor control.

Idle

Idle blank time： Idle waiting time, unit: s. Used to prepare for the next motor startup.

Align

Align angle： Rotor alignment angle, unit: °.

Align Id： Alignment current, unit: A. Corresponds to the Iq reference value in this FOC example code.

Align duration： Alignment duration, unit: s.

Ramp

Ramp Iq target： Open-loop forced current, unit: A.

Ramp ω target： End switching angular frequency at the transition to closed-loop control, unit: Hz.

Ramp duration： Total duration of open-loop ramp-up operation, unit: s.

Transition

Switch θ_err： Maximum allowable angle error during the transition from open-loop to closed-loop control, unit: °. This angle error refers to the difference between the open-loop forced angle and the angle estimated by the SMO during the transition.

Switch Min Iq： Minimum limit value of Iq during the transition from open-loop to closed-loop control, unit: A. The principle is to reduce the Iq reference value to minimize the difference between the open-loop forced angle and the estimated angle.

Observer hold count： Number of consecutive times the angle error meets the closed-loop switching condition. When the angle error is less than the set value of **Switch θ_err**, the counter increases by 1. If the accumulated count reaches the set value of **Observer hold count**, the system automatically switches to closed-loop control.

Switch time-out： transition timeout, unit: s. If the transition is not successful within the set **Switch time-out** duration, the system will be forced to the closed-loop control.

Speed-loop hold time： Duration of torque closed-loop control, unit: s. Before entering speed closed-loop control, the system executes torque closed-loop control for a period determined by the **Speed-loop hold time** setting.

Example

Figure 25 Parameters Configuration Example

By referring to the configuration parameters shown above and correlating them with the measured motor phase current waveforms, the practical significance of these parameters during operation can be more clearly understood.

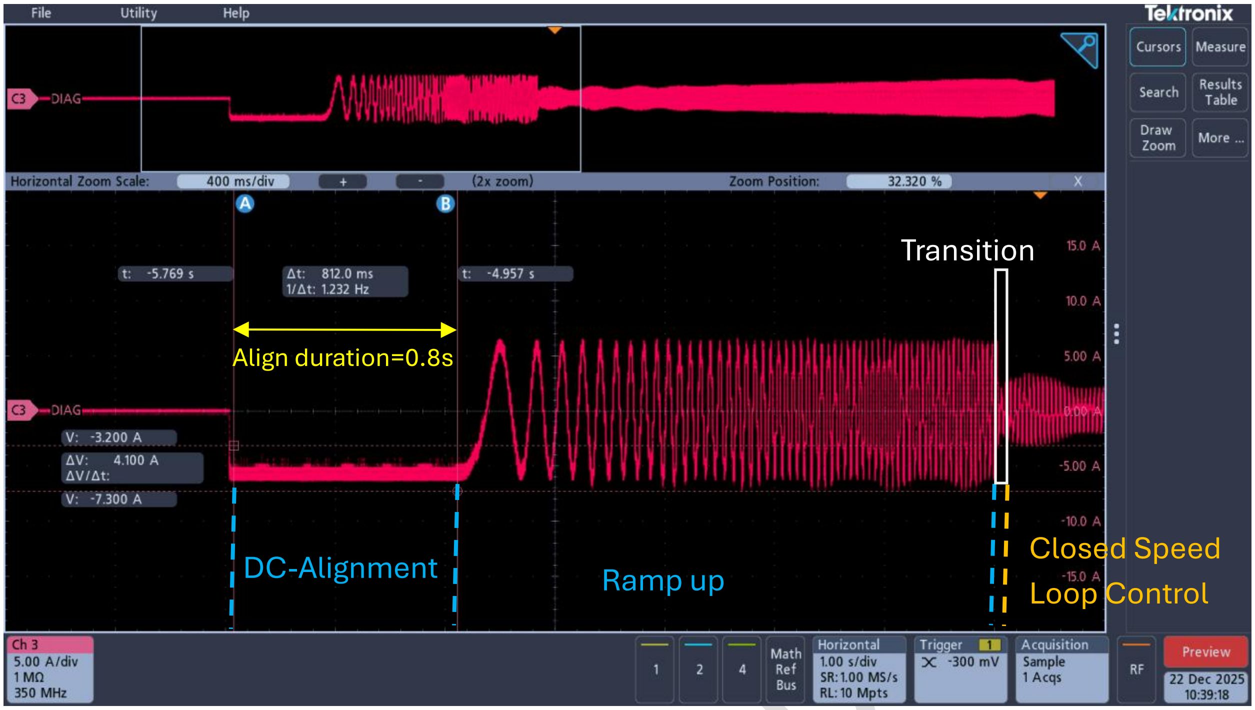


Figure 26 Align duration = 0.8s

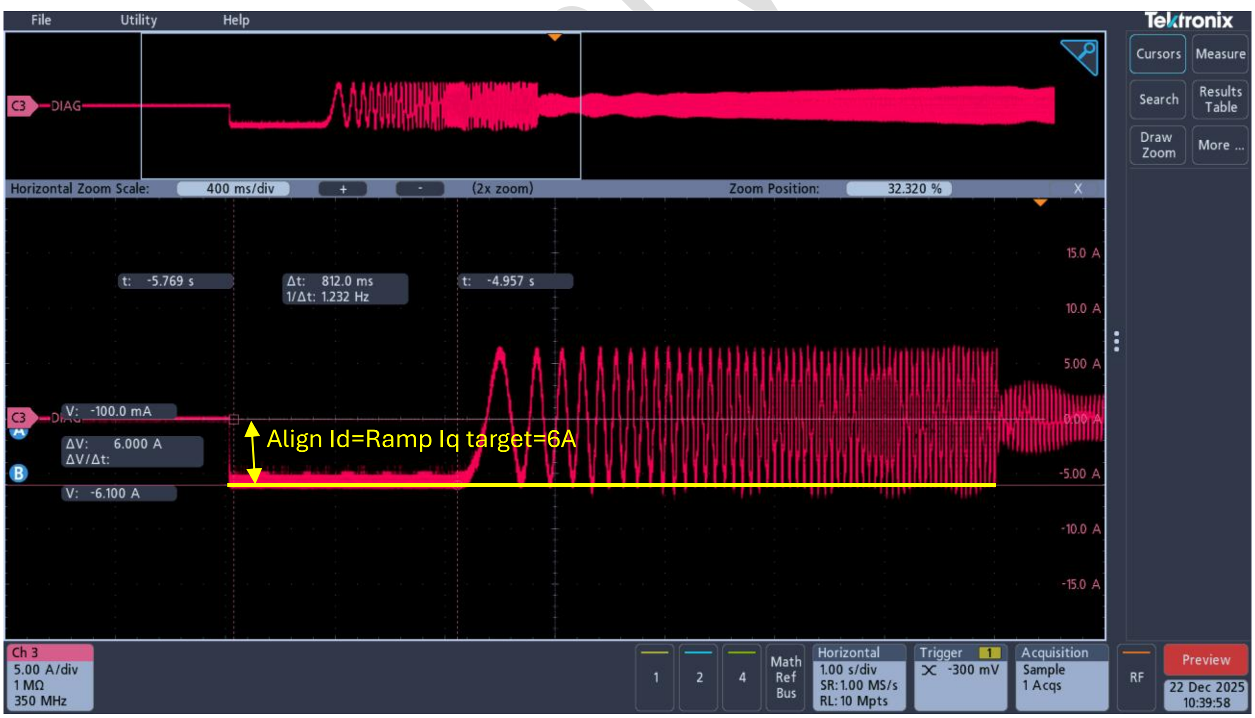


Figure 27 Align Id = Ramp Iq target = 6A

Figure 28 Ramp ω target = 60Hz

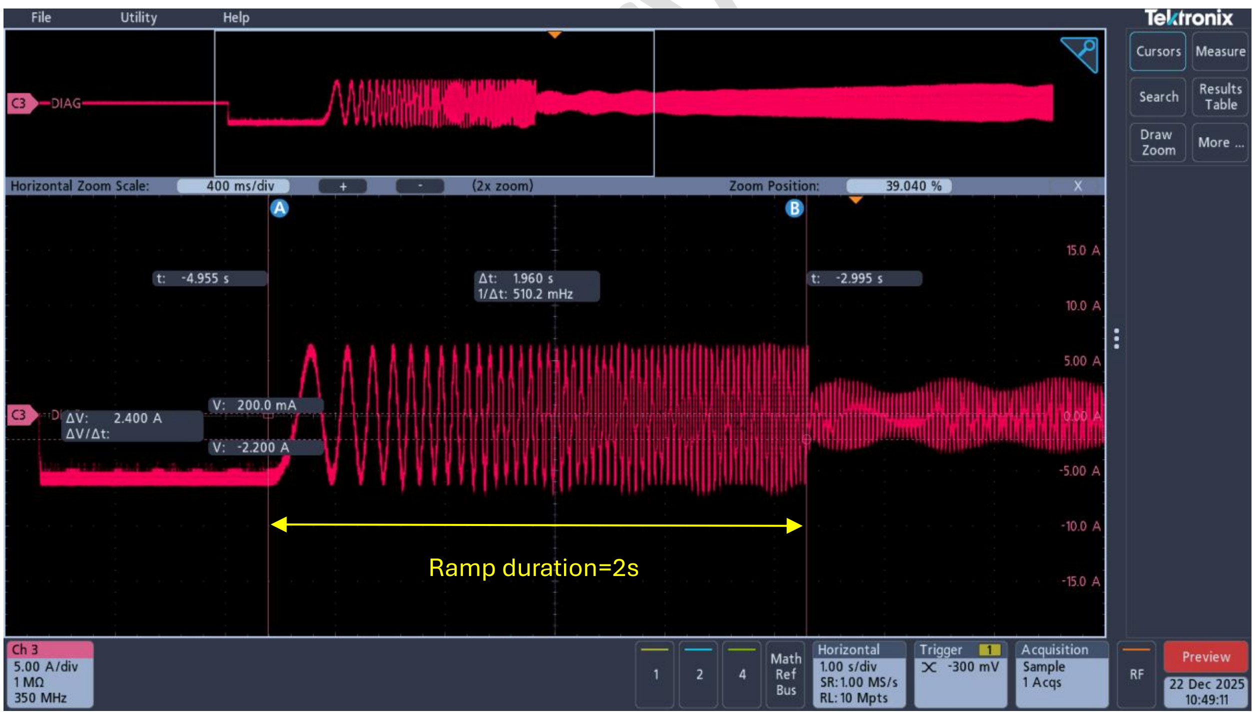
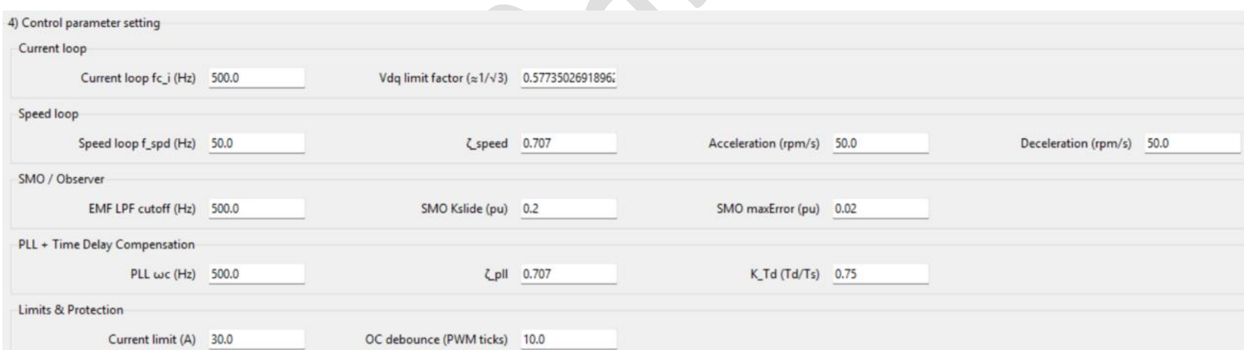


Figure 29 Ramp duration = 2s

Figure 30 Switch Min Iq = 2A

6) Control parameter setting



This section is primarily used to configure parameters related to the current-loop PI controller, speed-loop PI controller, SMO, PLL PI controller, and software over current protection. Among these, the proportional and integral coefficients of the current-loop, speed-loop, and PLL PI controllers are automatically calculated based on the input parameters in conjunction with the motor parameters. For users who want to review the detailed calculation formulas, please refer to **UserParameters_Guideline.txt**.

<u>Current loop</u>

Current loop fc_i： Current-loop cutoff frequency, unit: Hz. A higher cutoff frequency results in a wider system bandwidth and faster dynamic response, but may reduce system stability. The parameter should be tuned according to the actual application.

Vdq limit factor： Maximum limit values of V_d and V_q during SVPWM modulation under non-over modulation conditions. This parameter usually does not require modification.

Speed loop

Speed loop f_spd： Speed-loop cutoff frequency, unit: Hz. Typically 5 – 10 times lower than the current-loop cutoff frequency.

$\boldsymbol{\zeta\_speed}$： Damping coefficient, typical value: 0.707.

Acceleration： Acceleration, unit: rpm/s.

Deceleration： Deceleration, unit: rpm/s.

SMO/Observer

EMF LPF cutoff： Digital low-pass filter cutoff frequency, used to filter the back-EMF signal estimated by the SMO observer. A higher cutoff frequency results in poorer filtering but faster response; conversely, a lower cutoff frequency provides better filtering and smoother back-EMF waveforms. Typical values: when the current-loop execution frequency is 10 kHz – 40 kHz, the EMF LPF cutoff is recommended to be set within the range of 200 – 1000 Hz. The specific parameter should be tuned according to the actual application.

SMO Kslide： Gain coefficient of the SMO, expressed in per-unit values, with a range of 0 – 1.0. The recommended initial value is around 0.3, but the exact parameter should be adjusted based on the application.

SMO maxError： Maximum allowable current error for the SMO, expressed in per-unit values, with a range of 0 – 1.0. This is typically a small positive number. The recommended initial value is around 0.02, but the exact parameter should be tuned according to the application.

PLL + Time Delay Compensation

PLL $\boldsymbol{\omega_c}$： Phase-locked loop cutoff frequency, unit: Hz. The recommended initial value is set to the motor's maximum electrical frequency. The specific parameter should be tuned according to the actual application.

$\boldsymbol{\zeta\_pll}$： Damping coefficient, typical value: 0.707.

K_Td: Compensation factor for system delay, including ADC sampling and conversion delay, back-EMF digital filtering delay, code execution delay, and PWM update delay. Range: 0 – 1.0. The recommended initial value is within 0.6 – 1.0.

<u>Limits&Protection</u>

Current limit: Software over current protection threshold, unit: A. This parameter should be set according to the maximum operating current of the motor, and its maximum value must not exceed the effective maximum absolute current I_{abs_max} of the motor control demo board.

For the A89201 General Motor Control Demo Board:

$$I_{abs\_max} = \frac{1.2}{\text{G\_amp gain} \times \text{R\_shunt}}$$

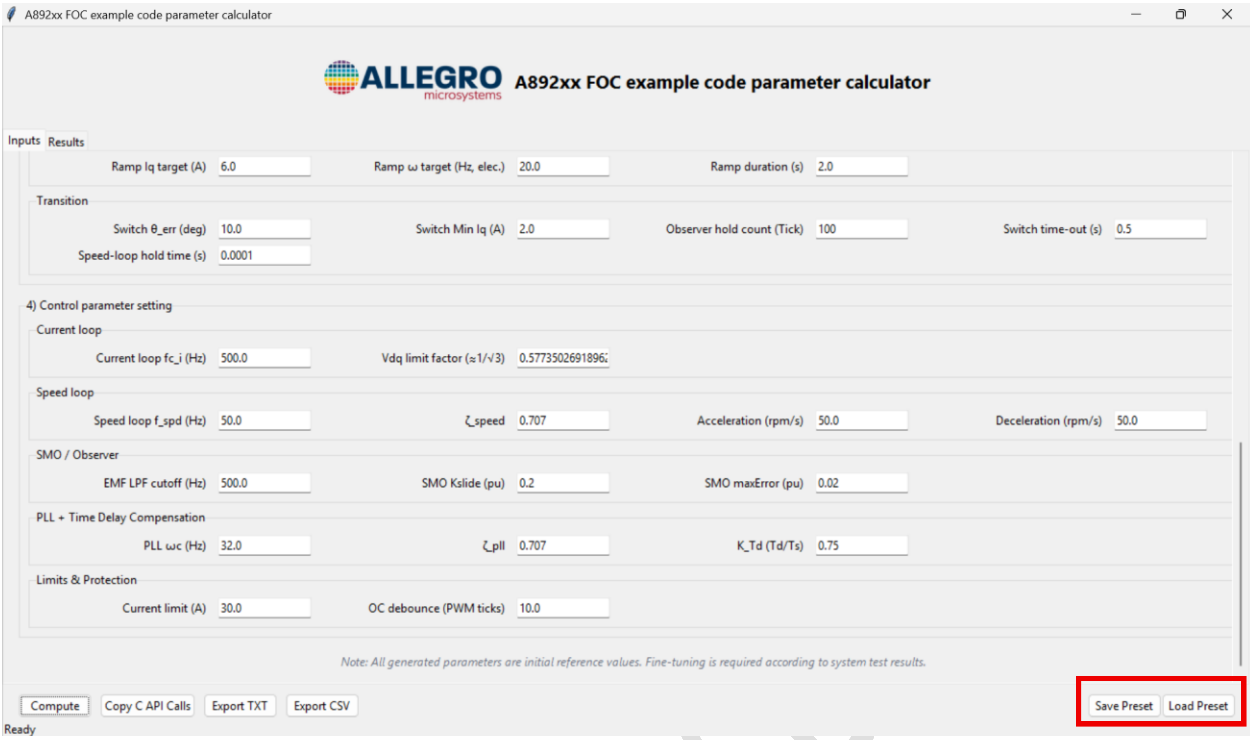
For the A89211/12/24 General Motor Control Demo Board:

$$I_{abs\_max} = min\left(\frac{1.2 - \text{Pedestal AMP Offset}}{\text{Sense AMP Gain} \times \text{R\_shunt}}, \quad \frac{\text{Pedestal AMP Offset} - 0.3}{\text{Sense AMP Gain} \times \text{R\_shunt}}\right)$$

OC debounce: Consecutive over current count. Each time the peak motor phase current exceeds the **Current limit**, the counter increases by 1. When the accumulated count reaches or exceeds the **OC debounce** setting, software over current protection is triggered (PWM output is disabled, control parameters and the motor control state machine are reset).

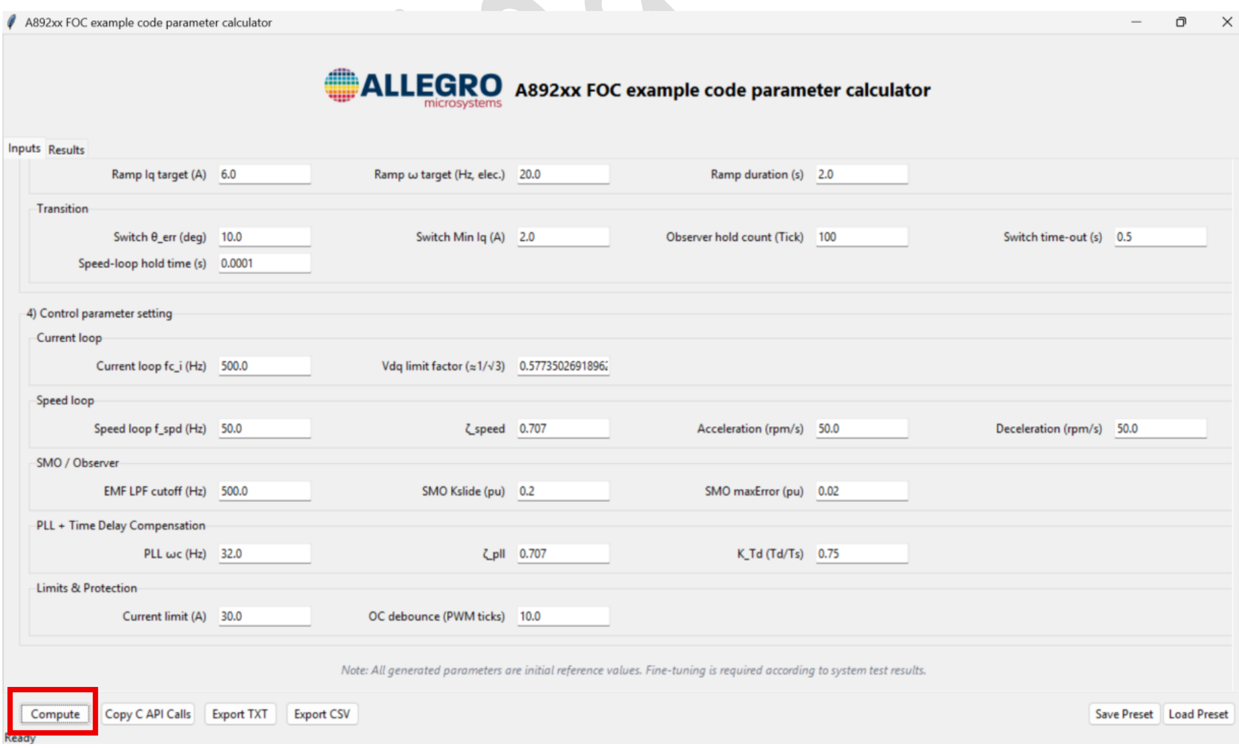
7) Save/Load Preset

After completing all the above control parameter configurations, click the **"Save Preset"** button to save the current settings for future use. Click the **"Load Preset"** button to load the previously saved configuration parameters.

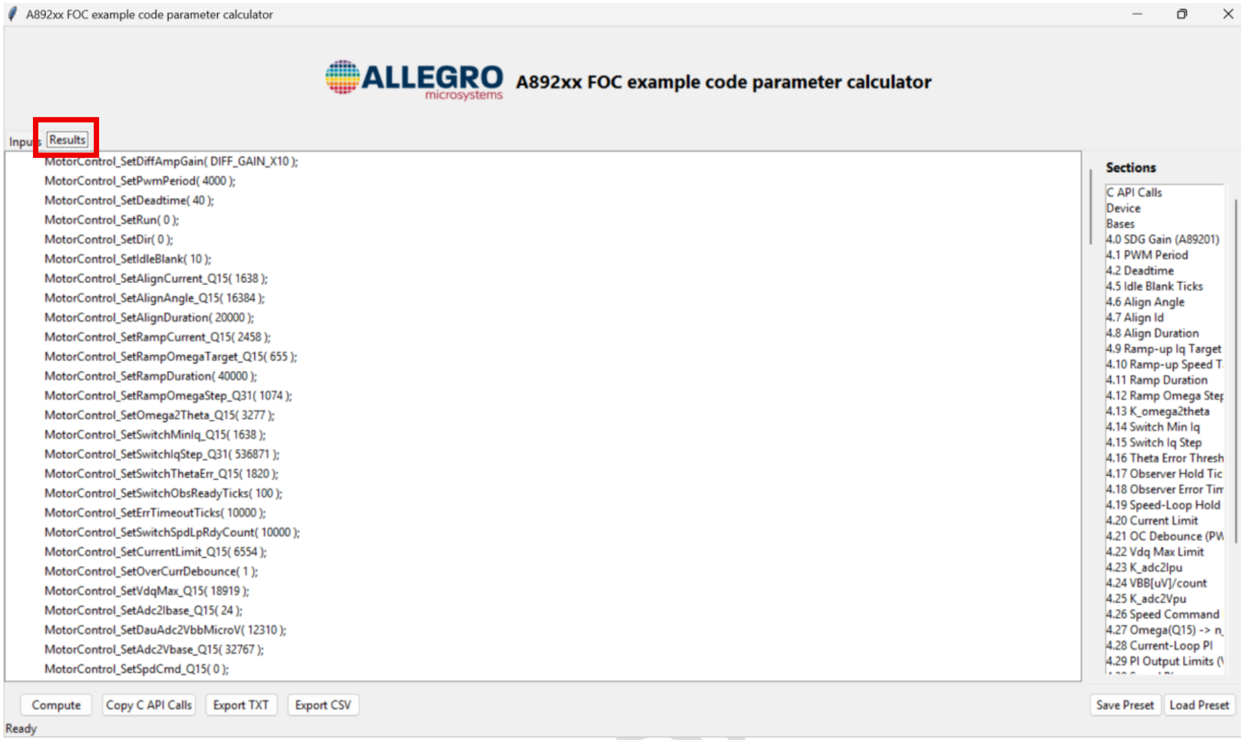


8) Generate user parameters for the FOC example code use

Click the **"Compute"** button shown below to generate user parameters that can be directly applied in this FOC example code.

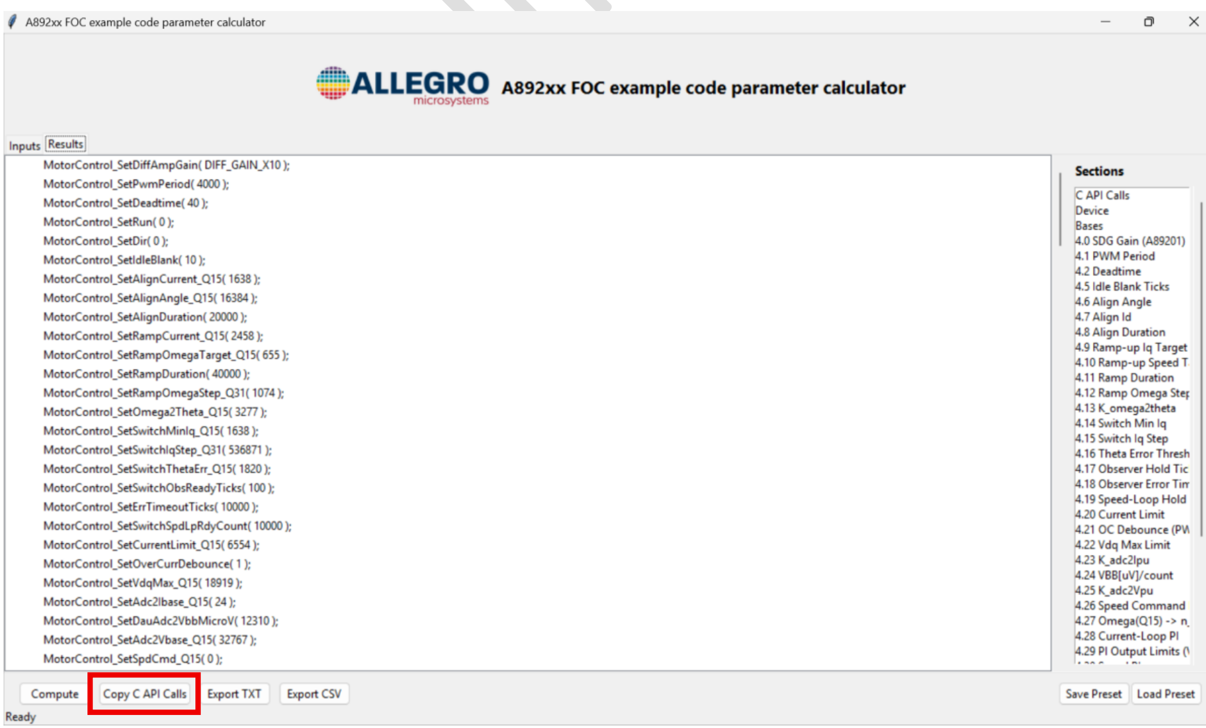


Click the **"Results"** button shown below to view the generated user parameters.



9) Load user parameters into the FOC example code

Click the "Copy C API Calls" button shown below to copy all the generated user parameters.



Then, load the FOC example code in the Keil, and paste the previously copied user parameters into **UserParameters.c** file at the location shown below.

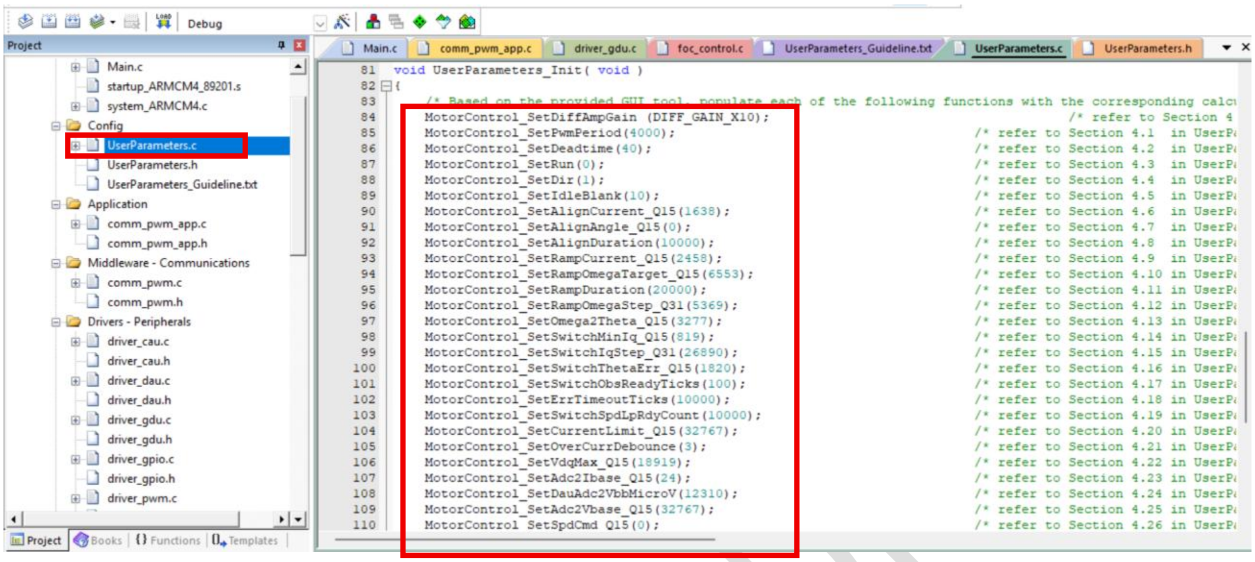


Figure 31 Load User Parameters into the FOC Example Code

6.3 J-Scope Waveform Visualization Tool

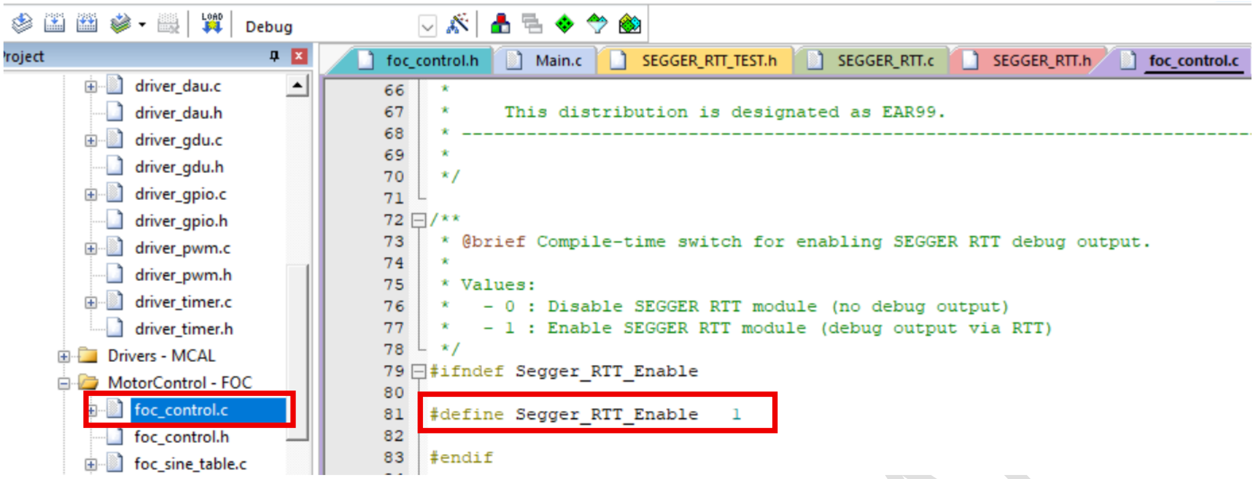
1. Introduction

SEGGER J-Scope is a waveform visualization software designed for embedded system debugging. It primarily uploads data from the microcontroller to the host PC via RTT mode and displays it in waveform form. RTT mode is implemented based on SEGGER's RTT (Real-Time Transfer) technology, requiring no additional hardware resources and without affecting the normal operation of the target board.

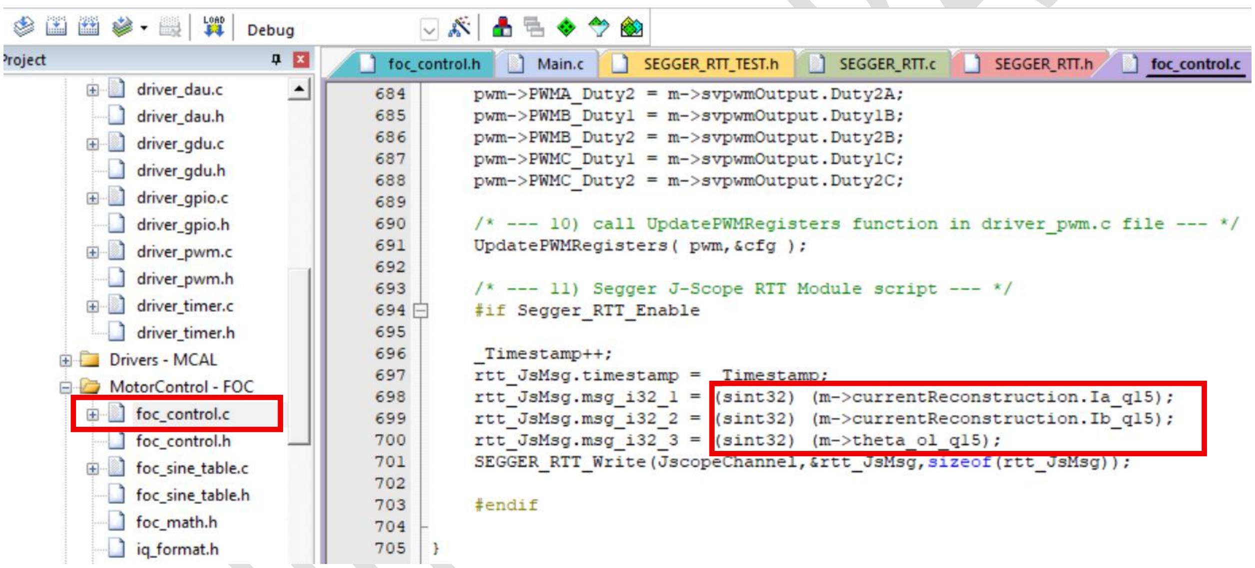
It should be noted that J-Scope must be used together with a debugger that supports RTT, such as J-Link BASE (V9/V10 and later versions). With J-Scope, users can efficiently obtain real-time data during motor operation, significantly reducing the difficulty of motor control system debugging and improving overall debugging efficiency.

Before using J-Scope in RTT mode for project debugging, the RTT function in the FOC example code must first be enabled and properly configured. The specific steps are as follows. Since the FOC example codes for A89201 and A89211/12/24 are similar, the following example is based on the A89211/12/24 code.

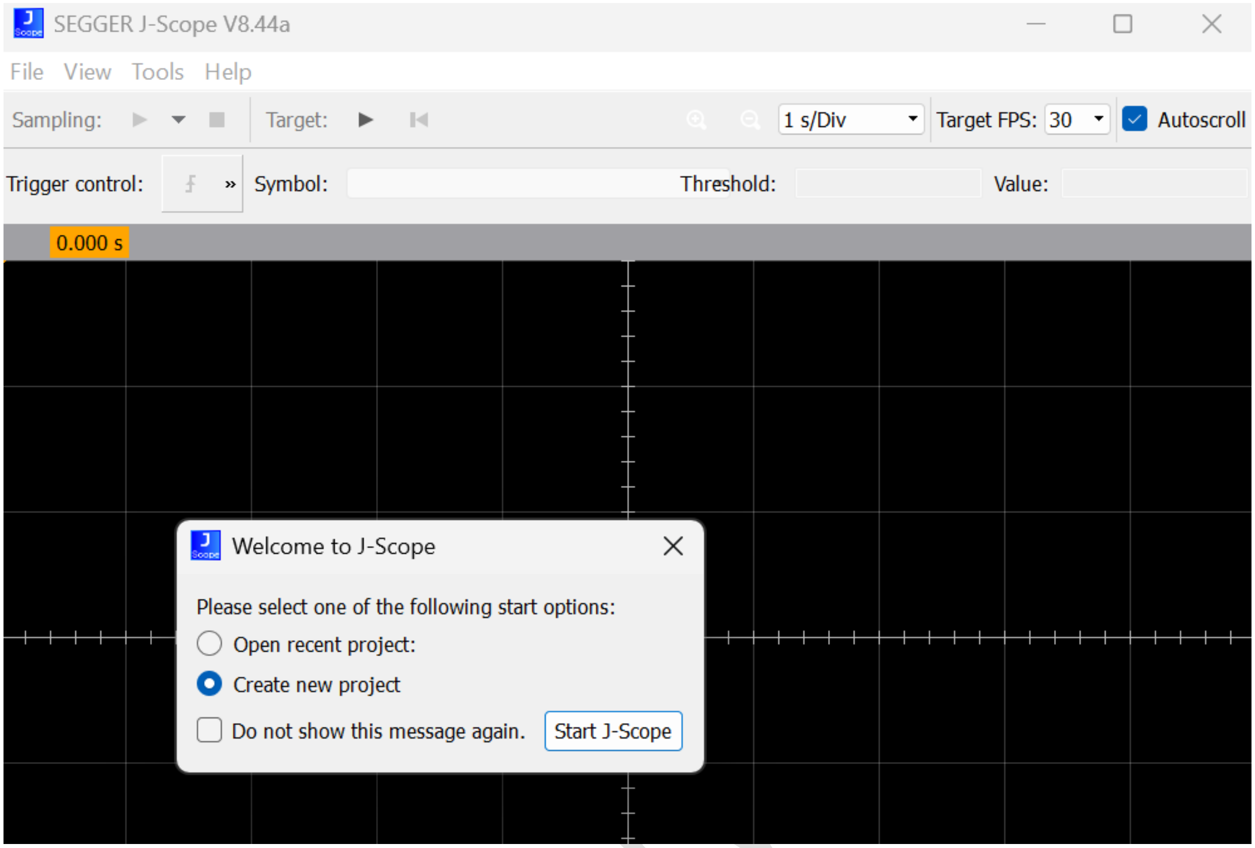
2. Operating Steps
    1) After opening the **foc_control.c** file, set the macro definition **Segger_RTT_Enable** as **1**.



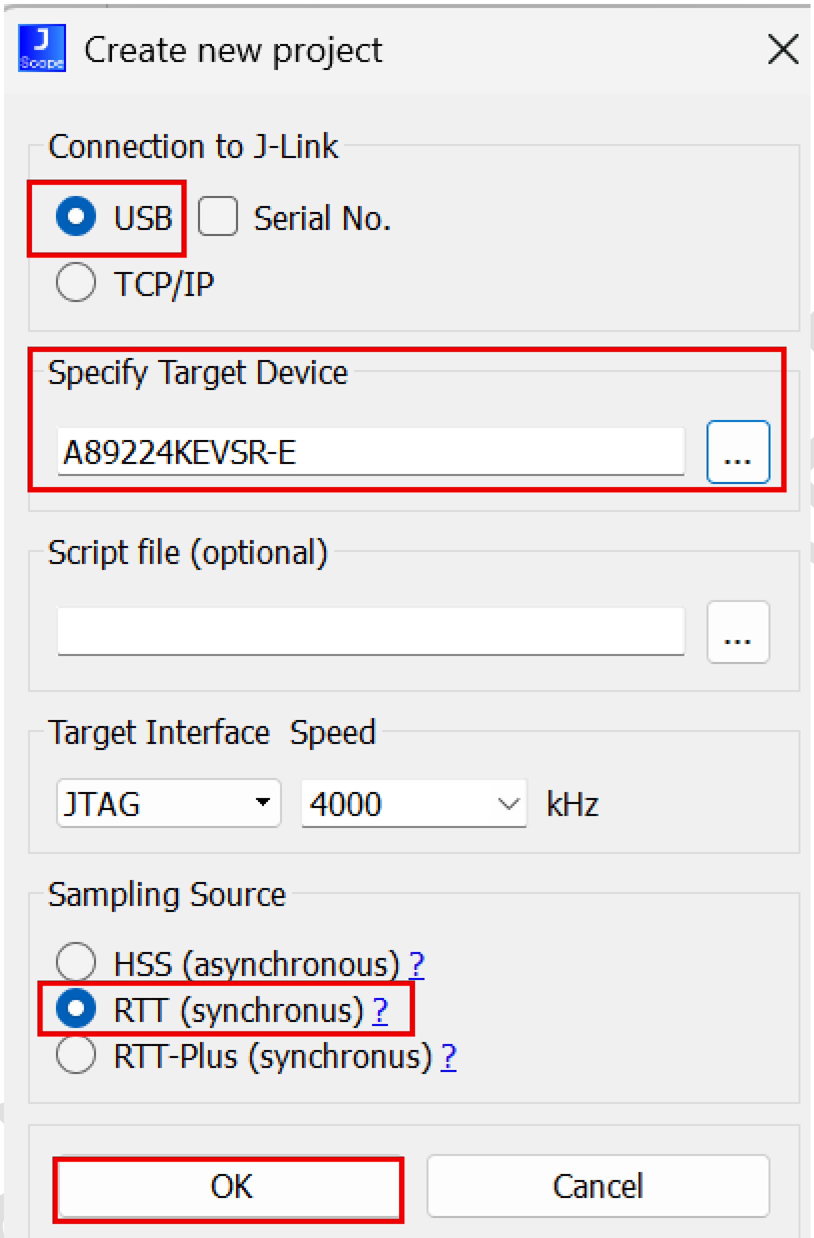
2) Enter the variables to be observed at the location shown in the figure below.



3) Launch the **J-Scope** and the configuration interface pops up. Check **"Create new project"**, then click **"Start J-Scope"**.



4) Configure the settings as shown in the figure below. If debugging the A89201 project, change **"Specify Target Device"** from **"A89224KEVSR-E"** to **"A89201KEVSR-C"**, while keeping all other configurations unchanged. After completing the configuration, click **"OK"**.



5) Click the Start Sampling button, as shown in the figure below.

6) After starting the motor operation, the variables to be observed will be displayed in waveform form, as shown in the figure below.

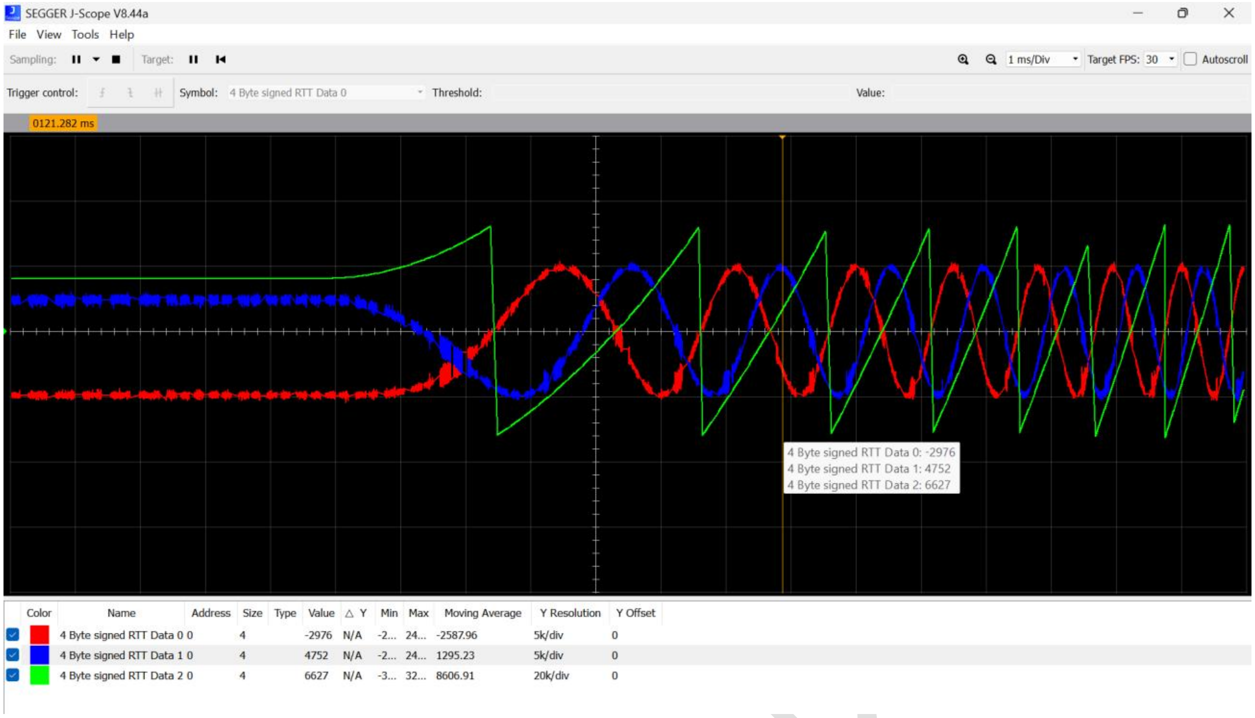
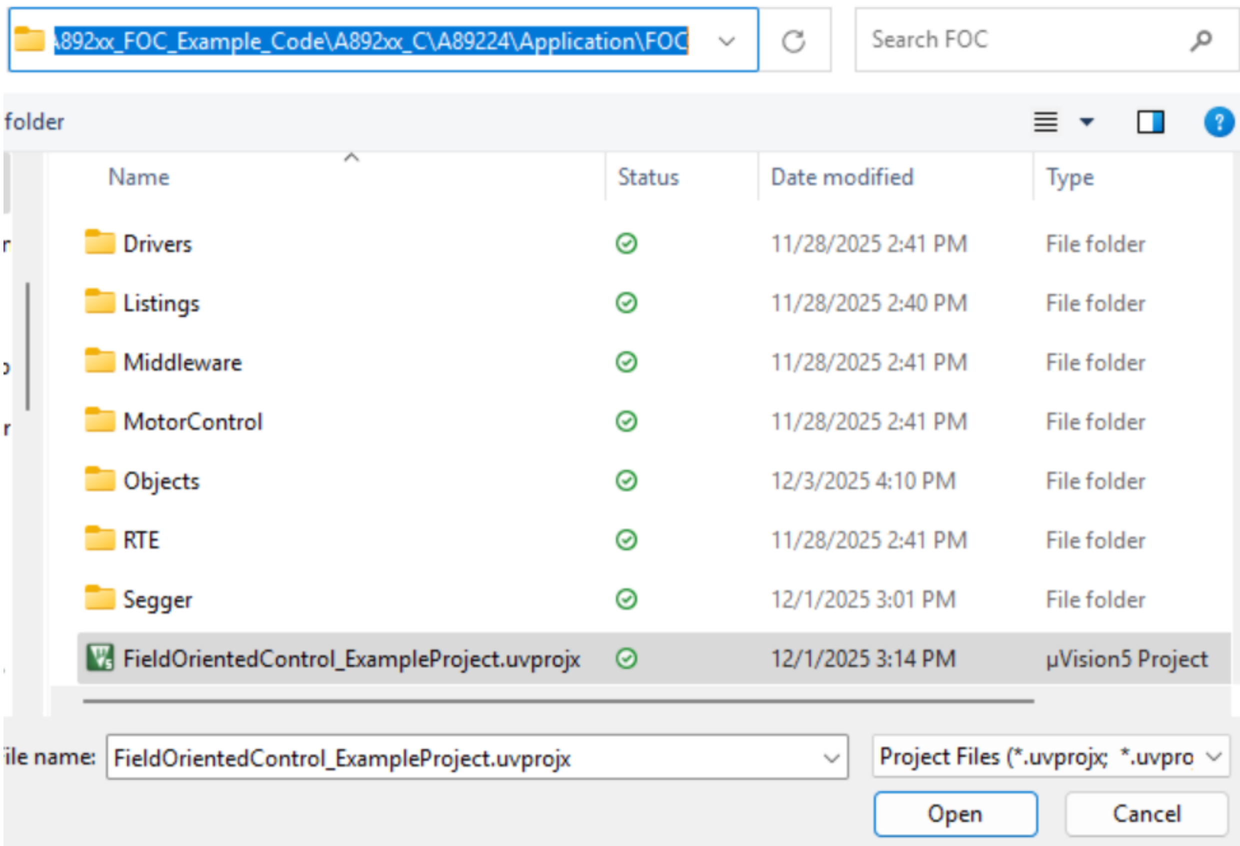


Figure 32 J-Scope Waveform Visualization

7 Tuning Steps

1. Following the instructions in the **"Wiring Guide"**, correctly connect the DC power supply, the three-phase permanent magnet synchronous motor, the PWM signal generator, and the ARM debugger. During this process, ensure that both the DC power supply and the PWM signal generator remain turned off.
2. Launch the Keil development tool and load the FOC example project file **FieldOrientedControl_ExampleProject.uvprojx**.



3. Use the **Allegro Parameter Calculation and Conversion Tool** (**A892xx_FOC_Example_Code_Parameters_Calculator.exe**) to configure the motor control parameters and generate user parameters that can be directly applied to the FOC example code.
4. Copy and paste the generated user parameters into the **"UserParameters.c"** file. The detailed operation steps have already been explained earlier and will not be repeated here.
5. After confirming that the applied supply voltage does not exceed the maximum safe operating voltage range specified for the A89201 or A89211/12/24 general motor control demo board, turn on the output of the DC power supply.

6. Compile the project and flash the code to the motor control demo board. After the flashing process is completed, turn off the output of the DC power supply.
7. After powering on again, enable the output of the PWM signal generator. Before that, please ensure the PWM signal amplitude is less than 5 V. The recommended signal frequency range is 100 Hz to 100 kHz. The valid PWM duty cycle range is 4% to 89% (default), corresponding to 5% to 100% (default) of the motor's maximum electrical frequency. When the PWM duty cycle is greater than 4%, the motor will start running; otherwise, the motor will stop.
8. Fine-tune the motor control parameters based on dynamic performance, repeating Steps 3 – 7 until the operating results satisfy application requirements.

Revision History

Number	Date	Description	Responsibility
–	January 29, 2026	Initial release	Jay Wang Bo Shao

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